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AI-Driven Analysis of Political Bias in Financial Media:

Impact on Retail Investor Behavior and Stock Market Dynamics

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Declaration of Authorship

“I hereby declare that this thesis, titled "AI-Driven Analysis of Political Bias in Financial Media: Impact on Retail Investor Behavior and Stock Market Dynamics", submitted as partial fulfillment of the requirements for the degree of MSc Data Analytics for Business at the KEDGE Business School, is my original work and has not been submitted previously for a degree or diploma at any other educational institution.

I confirm that:

1. I have fully acknowledge and reference all sources of information and data used in the preparation of this thesis.
2. All verbatim extracts have been distinguished by quotation marks and the sources of information specifically acknowledged.
3. No material other than that referenced and acknowledged has been used in the preparation of this thesis.
4. The work and research presented in this thesis have been carried out in accordance with the academic and ethical standards of KEDGE Business School.

I am aware of the consequences of breaching academic integrity policies of the KEDGE Business School, including the potential disqualification from the degree program.”

Signed,

A handwritten signature in black ink that reads "Daniel Rodriguez". The signature is written in a cursive style with a large, looped initial 'D'.

Acknowledgements

First and foremost, I offer my deepest gratitude to God for granting me the wisdom and strength to undertake this journey. His enduring grace has been a guiding light through every challenge, making this achievement possible.

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Deep appreciation is owed to Colombia and particularly to COLFUTURO because their scholarship program delivered the vital financial base that allowed these studies to proceed. The technical architecture found here stands on the shoulders of the open source community where specific libraries like TextBlob or VADER and Scikit Learn became absolute necessities. Transparency and reproducibility in this project depend entirely on those tools.

To my parents, thank you for your unwavering support and the countless sacrifices you made to get me to this point in life. You are the foundation upon which this work stands, without your example of hard work, I wouldn't be as motivated as I am today. To my girlfriend, Laura, thank you for your endless patience, your encouragement during the long nights of writing, and for keeping me grounded throughout this demanding process.

Finally, this thesis marks the end of one chapter and the beginning of another. It has reinforced my conviction that in a world increasingly filled with noise, the pursuit of objective truth, whether in data or in life, is the most worthy endeavor of all.

Abstract / Executive Summary

Real-time information flows drive modern asset pricing, yet the specific distortions introduced by media bias are difficult to quantify. While most news-sentiment pipelines treat publishers as ideologically neutral, this thesis tests whether publisher-level bias labels really add measurable signals beyond technical baselines. By integrating a novel dataset of high-frequency financial news, Reddit discussion threads, and market data for 10 large-cap tickers drawn from the S&P 500 and NASDAQ (2024-2025 sample window), this study constructs a reproducible machine learning pipeline to isolate the predictive value of behavioral signals.

Here Natural Language Processing (NLP) acts as the core mechanism to engineer distinct behavioral features, TextBlob parses professional news coverage while VADER extracts ideological cues from social media. A strict time series cross validation framework subsequently tests these inputs by employing Logistic Regression alongside Random Forest or Gradient Boosting classifiers. And that is how the results demonstrate that behavioral features provide a reliable predictive advantage over technical baselines since Logistic Regression performed best among the models to achieve an AUC of 0.523. This metric represents a tangible lift of 0.023 above naive benchmarks. A hierarchy of influence appears within the feature analysis because Retail Sentiment functions as the primary driver with a standardized coefficient of 0.108 that far exceeds the 0.002 score associated with professional news sentiment. So, political bias proved to be another material factor since average bias captured ideological direction and scored 0.036 to notably outperform Bias Dispersion.

Also more specific granular views highlight differences at the source level where left leaning outlets achieved a directional accuracy of 52.1% compared to the 46.8% accuracy seen in Centrist outlets. Such findings challenge the traditional reliance on neutral news sentiment because retail activity generates the strongest discrimination while political bias features add a consistent and interpretable layer to the signal.

This thesis ultimately contributes a reproducible framework for embedding political nuance into AI driven financial forecasting. Given the modest but statistically significant lift (+2.3%), practitioners are advised to treat these signals as auxiliary filters within a broader risk

management framework rather than standalone alpha generators, subject to rigorous transaction-cost analysis.

*All code and data used in this study are available in the following repository:
https://github.com/daniel-st3/Daniel_Rodriguez_MSc_Thesis_Final*

Keywords: Financial Sentiment Analysis, Political Bias, Natural Language Processing, Retail Investor Behavior, Efficient Market Hypothesis, Algorithmic Trading, Machine Learning, Behavioral Finance.

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Glossary of Terms

AUC (Area Under the Receiver Operating Characteristic Curve) The Area Under the Receiver Operating Characteristic Curve serves as the primary metric for distinguishing between upward and downward market moves. A score of 0.5 in this context indicates a performance level equivalent to random guessing.

Bagging (Bootstrap Aggregating) Bagging or Bootstrap Aggregating functions as a critical mechanism to reduce variance within the Random Forest classifier since this ensemble technique trains multiple decision trees on isolated subsets of the price and news data.

F1-Score This metric provides a single performance indicator by computing the harmonic mean of precision and recall. It effectively balances the trade off between false positives and false negatives.

Gradient Boosting Weak learners build sequentially in Gradient Boosting where each iteration attempts to correct the errors of previous trees. The algorithm showed signs of overfitting on this specific dataset despite its inclusion in the comparison.

Look-Ahead Bias Look ahead bias emerges as a critical failure state whenever future information accidentally bleeds into the training environment so strict time series cross validation becomes necessary to neutralize the threat. This method enforces a rigid chronological barrier that restricts the model to learning solely from past observations.

MACD (Moving Average Convergence Divergence) The relationship between the 12 day and 26 day exponential moving averages constitutes the Moving Average Convergence Divergence.

MBFC (Media Bias/Fact Check) Media Bias Fact Check serves as a third party proxy for establishing reputation. The methodology involves assigning labels such as Left or Right Center to the publisher itself rather than categorizing individual articles. This strategy acts as a heuristic that aids in the detection of systemic bias.

NLP (Natural Language Processing) Natural Language Processing sits at the intersection of linguistics and computer science. It allows for the quantification of unstructured text found in Reddit threads and financial news. The process transforms raw language into vector inputs like sentiment scores.

Precision Precision measures how accurate positive predictions actually are. It does this by taking the ratio of true positives and comparing it against the total volume of positive signals the model generated.

Random Forest Random Forest operates as an ensemble method designed to boost stability.

Recall (Sensitivity) Sensitivity or Recall indicates the ability to capture positive instances so a higher ratio of true positives to the total actual class means the model misses fewer significant signals.

RSI (Relative Strength Index) The Relative Strength Index evaluates the magnitude of recent price changes so traders use this momentum indicator to spot overbought or oversold market conditions.

Sentiment Analysis Sentiment analysis transforms subjective information into hard numbers.

TextBlob Processing professional news articles relied on TextBlob. This Python library provides a simple API for executing standard natural language tasks.

Time-Series Cross-Validation Look ahead bias is prevented by Time Series Cross Validation because this technique respects temporal order so that the training set contains only observations preceding the test set.

VADER VADER is specifically designed for the nuances of social media. The pipeline uses this lexicon and rule based tool to determine the polarity of Reddit discussions.

Introduction

Algorithmic trading relies on the assumption that data inputs are objective. However, financial news outlets filter corporate events through distinct editorial lenses, subtly framing how investors interpret risk. A central managerial dilemma arises from this subjectivity: should algorithms treat political ideology as noise to be ignored, or as a quantifiable risk factor?

Increasing polarization in the information landscape forces investors to make a binary choice. They can view news outlets as neutral conduits. Alternatively, models can incorporate the reputational prior of a publisher to filter noise and capture alpha. This research determines if political bias metrics add a statistical edge to predictive algorithms. The resulting signal must justify the implementation complexity. Bias affects the tone and framing of stories. It dictates the selection of news. These editorial choices alter how the market interprets data.

Retail participants currently endure the most significant market susceptibility because they lack the sophisticated hedging instruments required to mitigate sudden distortions. Institutional capital operates differently by adhering to rigid valuation frameworks while retail liquidity chases sentiment often manufactured by financial media narratives. Biased reporting structures the discourse in a way that frequently misaligns expectations so this divergence generates inefficiencies where short term price action severs its connection to intrinsic value.

Current academic work leaves this relationship mostly untouched since past inquiries tended to focus on the general weight of media sentiment or adhered strictly to fundamental analysis. This creates a noticeable void in the field because very few researchers have actually isolated political bias as an independent variable or investigated the direct fallout it has on market mechanics.

Understanding the mechanisms through which political bias in financial media affects retail investor sentiment is essential for improving market transparency and creating fairer trading environments. Refining the predictive power of AI-driven financial models contributes to a more democratized understanding of market mechanics for everyday traders. This thesis seeks to explore how the political leanings of major financial news outlets shape investors' perceptions and drive short-term movements in stock prices. By examining whether different forms of media

coverage alter the framing of company-related information, it aims to determine if retail trader sentiment is swayed and subsequently reflected in trading activity and stock performance.

Contextual Background

The democratization of market access changed the financial landscape forever over the last decade. Equity markets used to be the private territory of institutional giants like hedge funds or investment banks. These players leveraged proprietary data terminals to secure a massive advantage. Information asymmetry defined the entire era. Professional analysts accessed earnings reports early so prices moved long before retail capital could react. Those barriers finally crumbled after 2020. Zero commission platforms such as Robinhood and eToro slashed entry costs down to absolutely nothing. Social media simultaneously emerged as a potent financial news terminal where Reddit communities or YouTube influencers effectively replaced traditional analyst reports. These channels now serve as the primary due diligence mechanism for millions of new entrants so this structural evolution ultimately inserted viral sentiment as a critical variable in asset pricing.

Institutional capital generally adheres to fundamental valuation models like discounted cash flow, but retail capital frequently chases momentum and narrative framing. The 2021 GameStop squeeze demonstrated that coordinated retail sentiment possesses the power to override fundamental valuations. It forced institutions to abandon short positions. Information framing subsequently drives capital flow within this modern ecosystem. A news story depicted as a catastrophic failure moves markets differently than one presented as a strategic buying opportunity. Consequently, understanding the political and emotional tint of financial news is no longer just a sociological exercise; it is a financial necessity for predicting short-term volatility.

The Attention Economy and Media Polarization

Alongside the rise of the retail investor is the fragmentation of the media ecosystem. The traditional model of objective, neutral financial reporting (e.g., the Associated Press style) competes with an attention economy that rewards sensationalism and polarization. Media outlets,

driven by ad-revenue models dependent on clicks and engagement, have an incentive to frame financial news through emotionally charged or politically aligned lenses.

Media polarization fundamentally distorts the reality perceived by the retail investor because a labor strike at a major automaker illustrates this dynamic clearly. One outlet might frame the event as a necessary step for workers rights that ensures long term stability while a competing source depicts the exact same strike as a disaster for competitiveness. These conflicting narratives signal immediate sell pressure to one group and resilience to another meaning that investors segregated into these ideological echo chambers inevitably trade on divergent information. Such behaviors leave observable footprints in the market data that machine learning models can detect to secure a predictive advantage.

The Role of Alternative Data in Modern Finance

Incorporating non-traditional datasets into investment strategies represents one of the most significant paradigm shifts in modern financial history. For decades the dominant analytical framework relied exclusively on structured data which includes standard financial statements, price history, and trading volume. This structured approach assumed that all relevant information was contained within the numerical records released by public companies and exchanges. However the digital revolution has generated an explosion of unstructured information that exists outside these traditional channels. This new category of information is known as alternative data. This type of data includes diverse non-market signals, data that goes from satellite imagery to credit card logs in this case the most pertinent to this study are the domain of web-scraped textual data from news and social media.

The rise of alternative data is driven by the realization that alpha or excess return is increasingly difficult to generate using standard datasets that are commoditized and instantly priced in by high frequency algorithms. In this hyper competitive environment the edge belongs to market participants who can decode the signals hidden in unstructured noise.

Textual data stands out as the largest and most complex component of the alternative data universe. The market contends with a daily deluge of analyst reports, news articles, and social media posts, creating a continuous stream of qualitative information that actively steers market psychology. Quarterly earnings reports provide a static snapshot of past performance, whereas textual data offers a dynamic pulse of investor sentiment as narratives form. Utilizing this resource presents a difficulty. The data is voluminous and linguistically complex. Regression models cannot ingest the raw text directly. The analysis requires rigorous preprocessing to convert this unstructured input into quantified metrics.

This thesis contributes to this shift toward alternative data. By transforming the qualitative concepts of political bias and retail sentiment into quantitative time series vectors it demonstrates how unstructured text can be operationalized. This process involves more than counting words. It requires a deep understanding of the semantic context and the sociological source of the information. The use of Natural Language Processing or NLP bridges the gap between human language and mathematical modeling. It turns the subjective bias of a journalist or the emotional outburst of a retail trader into a structured feature that a machine learning algorithm can interpret and utilize for short-term forecasting. This transition from intuition to computation is the defining characteristic of the current era of quantitative finance.

The Retail-Institutional Disconnect

The market anomalies which surfaced post 2020 revealed a fatal blind spot within traditional asset pricing frameworks. Institutional algorithms had historically relegated retail order flow to the status of mere noise or simple liquidity provision. This entire logic rested on a fragile assumption that retail capital would forever remain fragmented yet that premise disintegrated during the meme stock volatility event. Coordinated social sentiment managed to overwhelm fundamental valuation mechanics. Funds that neglected the ideological cohesion of these new participants subsequently encountered severe liquidity crises. The disaster occurred not because the fundamental models were mathematically flawed but simply because they attempted to function without essential behavioral inputs.

The challenge for quantitative finance is no longer just accessing data but decoding the intent behind it. A generic sentiment score of "negative" does not capture whether a price drop is driven by a disappointing earnings report or a politically motivated boycott. These two drivers decay at different rates and require different hedging strategies. By flattening all news into simple polarity scores, current NLP pipelines discard the "reputational prior" of the publisher. A critique from a politically aligned outlet often carries a different weight in the mind of a retail trader than the same critique from a wire service. This thesis argues that the political bias of the source is not merely a metadata tag but a fundamental filter that determines how information converts into price action.

The Gap in Explainability

Implementing "black box" neural networks to predict these movements solves the prediction problem but fails the risk management mandate. A portfolio manager cannot deploy capital based on a model that outputs a probability without a thesis. If one cannot explain why a bias signal predicts a price reversal, we cannot trust the signal during a regime shift. This research specifically avoids complex "deep" architectures in favor of interpretable feature engineering. Existing risk models could use an upgrade instead of a complete overhaul so this analysis isolates the unique predictive value of political bias to fill that specific gap. The study investigates the idea that the political slant found in news coverage functions as a measurable risk factor because this specific element operates in a way that is very similar to common metrics like momentum or volatility.

Research Question

How does the influence of political bias in major financial news outlets on retail trader sentiment improve short-term stock price predictions through NLP models?

(Note: In this context, "influence" is operationally defined as the incremental contribution to predictive accuracy in a time-ordered machine learning setup. The study makes no causal claims regarding market mechanics.)

Justification

This research question arises from a pressing need to address the underexplored role of political bias in financial media narratives. While prior studies have demonstrated the impact of media sentiment on stock performance, they often fail to account for the underlying political leanings that shape these sentiments. By integrating political bias as a quantifiable factor, this study bridges a critical gap, offering a more nuanced understanding of how biased reporting affects investor behavior and market dynamics. The inclusion of AI and NLP models ensures that the analysis is both innovative and scalable, contributing to the growing body of research on AI-driven financial forecasting.

Focus

The analysis draws its core data from major financial news outlets that hold established political leanings such as The Wall Street Journal and The Guardian or The New York Times to capture a full spectrum of bias. This professional coverage pairs with retail trader sentiment taken directly from online discussions and brokerage feeds. Connecting these behavioral inputs to short term price movements highlights the direct impact of media bias since the market feels the effects immediately rather than theoretically.

The final dataset spans the period from June 2024 to June 2025 and consolidates 2570 ticker day observations into a single framework. This universe concentrates on 10 high volatility stocks from the S&P 500 and NASDAQ which includes major players like AAPL and NVDA or TSLA. A constrained set of tickers was chosen rather than a broad market index. This specific focus made the manual qualitative audits much easier to manage. It also ensured that bias labels mapped correctly to real world narratives.

Trading days remain balanced across the dataset but the data preserves the natural class imbalance of the market period where roughly 53% of samples represent upward movement. Data collection consisted of using the official Reddit API with a specific News API aggregator, of course, filtering strictly for ticker symbols was a requirement because this step ensured the signal remained clear and accurate throughout the entire workflow.

Literature Review

The connection between media narratives and investor sentiment is widely recognized, yet current models fail to capture the specific interaction between political framing and high-volatility events.

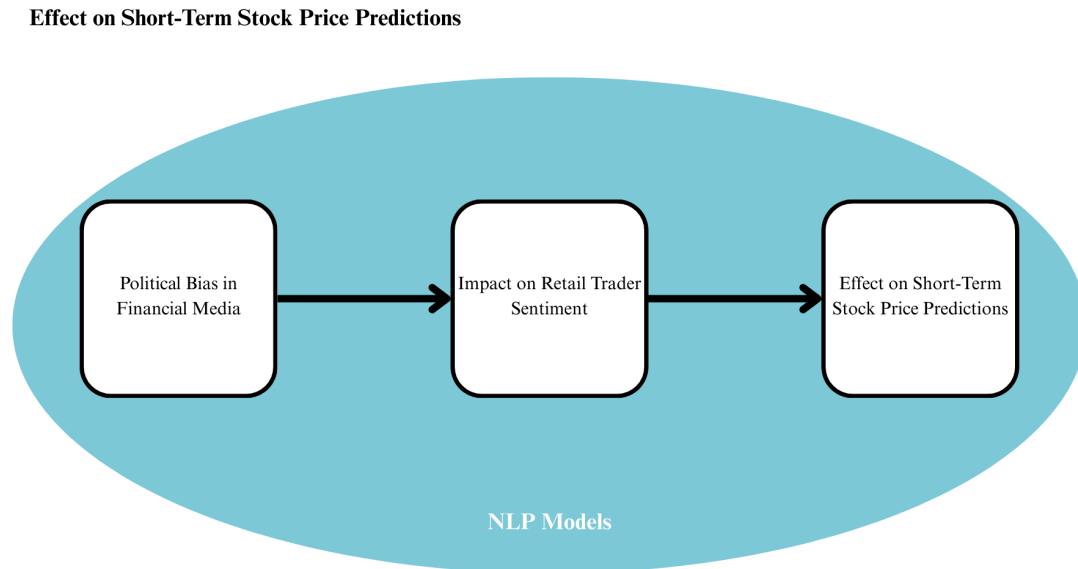
The sheer speed of financial news is relentless. The volume is just as overwhelming. This reality forces the adoption of advanced tools that can parse massive amounts of text. This capability is essential. Information flows like these often trigger severe market volatility. Old school methods frequently fail to capture the nuance locked inside these articles. They lean too heavily on structured numbers. They completely ignore the messy complexity of unstructured text.

Recognizing this, Artificial Intelligence (AI), particularly Natural Language Processing (NLP) techniques, has surged in the financial domain. Previous studies have explored the impact of media sentiment on financial markets (Kumbure et al., 2022; Goldman et al., 2024), but have seldom integrated political bias into their analyses. Sentiment analysis itself, a core NLP technique, aims to identify and classify emotions and opinions in text, facilitating the evaluation of attitudes expressed in written or spoken language. The core relationships explored in this thesis are illustrated in **Figure 1**.

This research seeks to fill that gap by evaluating the influence of political framing in financial news on retail investor sentiment and its effects on short-term market dynamics through NLP models. The following literature review synthesizes academic literature from peer-reviewed finance and computer science journals alongside select working papers and professional sources.

Figure 1

Conceptual Pathway of Political Bias, Sentiment, and Price Prediction



Note. Author's own conceptual diagram summarizing the thesis framework.

Theoretical Foundations

Fama (1970) established the theoretical standard for market efficiency by arguing that prices instantly discount every piece of public data. This framework implies that generating consistent alpha from news is mathematically impossible yet real world trading rarely adheres to such perfect rationality. Markets frequently demonstrate excess volatility and sudden reversals that contradict the logic of fundamental updating. Behavioral finance addresses these gaps by showing how investor psychology and narrative framing drive prices away from their intrinsic value. A key component of this deviation is what Black (1986) defined as noise or the phenomenon where trading volume is driven by signals that possess no actual relevance to the asset's health.

De Long et al. (1990) built upon this view by formalizing noise trader risk to show that when sentiment driven traders push prices away from fundamentals rational arbitrageurs often hesitate to correct the mispricing immediately. They fear the gap might widen before it eventually

converges so this logic of limited arbitrage explains why factors like retail discussion predict short term directionality even if they lack fundamental information.

Social transmission operates as a powerful secondary mechanism driven by the dual forces of herding and informational cascades because individuals frequently observe peer actions and rationally choose to follow the crowd. This choice is often made because they infer the group holds private information or simply find the cost of deviation too high. Banerjee (1992) proved herding happens even when agents possess private signals while Bikhchandani et al. (1992) described a similar phenomenon called informational cascades where early signals trigger widespread imitation. These dynamics create specific consequences for financial markets since media framing directs investor attention just before retail communities amplify those narratives through repetition and coordination.

In this thesis, these theoretical perspectives support the modeling choice to treat (i) retail sentiment as a measurable proxy of collective attention and noise, and (ii) publisher level political bias as a stable attribute that can influence framing and interpretation at the point of information exposure. Importantly, these frameworks are used to motivate predictive feature design under a time ordered evaluation setting, rather than to claim causality.

Media Bias and Market Sentiment

Media bias is widely known to shape public perception, a process that Steppat et al. (2022) attribute to the primary engines of media fragmentation and polarization. Individuals consequently seek out information that validates their existing ideological preferences. This mechanism heavily impacts the retail investor. It alters the interpretation of financial news. Confirmation bias intensifies and directly modifies investment decisions. Beyond simply reinforcing existing views, this selective exposure can lead to a distorted understanding of market realities. If an investor consistently consumes news from a source that downplays risks or inflates positive prospects for a particular sector aligned with their political leanings, they may develop an overly optimistic outlook, leading to miscalculations in risk assessment and portfolio allocation. Goldman et al. (2024) extend this perspective, demonstrating empirically that politically polarized financial news impacts market stability by altering the framing of economic

narratives. When news outlets with differing political leanings present conflicting accounts or interpretations of the same financial events, it can lead to confusion and disagreement among investors. This effect can be particularly pronounced for firms perceived as politically salient, where polarized reporting might signal increased trading volume as investors react to divergent narratives. Detecting this bias is critical for objective analysis, yet difficult because it often appears in subtle framing choices rather than explicit keywords. Simple keyword scanning fails to capture how facts are selectively highlighted or omitted by different outlets.

The reaction to framed news drives short term dynamics because the market prices major headlines almost instantly yet Salbrechter (2024) notes that residual predictive power often persists over shorter horizons despite this rapid absorption. Kerssens (2024) suggests that sentiment derived from news offers a latency advantage over traditional indicators but warns that speed becomes a liability if the underlying sentiment stems from biased reporting. Investors reacting to a distorted narrative inevitably drive prices away from fundamental value to generate volatility that persists until the market eventually corrects for the bias.

Sentiment Analysis and Predictive Models

Sentiment analysis operates as a specialized branch of NLP focused on identifying emotional tones, a capability that effectively transformed financial forecasting by enabling the integration of unstructured data like news articles and social media posts. Hasani (2024) emphasizes that pairing this analysis with AI significantly enhances predictive accuracy because modern algorithms can now detect subtle cues such as sarcasm or irony. Architectures like BERT and the GPT series extract these sentiments from diverse inputs, provided they undergo rigorous training within financial corpora. Kumbure et al. (2022) note that the technology bridges the gap between investor psychology and market trends.

While prior research has established a link between news sentiment and market volatility (Kumbure et al., 2022) these studies frequently treat media outlets as homogeneous pipelines of information. They often fail to account for the ideological framing effects where the same economic event is portrayed differently by Left vs. Right leaning sources that likely modulate

investor reaction. This study addresses that gap by treating publisher ideology not as noise but as a distinct informational signal.

Early sentiment analysis relied heavily on lexicon based tools like VADER and SentiWordNet because they utilized predefined dictionaries to assign polarity scores with high computational efficiency. This strategy nevertheless struggles when confronting the nuances of context or sarcasm and domain specific language. Modern methodologies have consequently pivoted toward advanced machine learning where supervised classifiers like Naïve Bayes and Random Forests train on labeled data to extract more reliable signals. Deep learning architectures provide even stronger mechanisms by leveraging attention layers within Transformer networks and recurrent models to parse long range dependencies. LSTM models excel specifically at managing the sequential nature of financial data while Transformers like BERT offer robust contextual understanding. Empirical validation for this shift appears in studies like Islam and Monira (2025) whose analysis of the Dhaka Stock Exchange demonstrates that ML techniques consistently outperform traditional forecasting. The subsequent development of domain specific models such as FinBERT has further refined the capacity to analyze complex financial text.

Cross-Cutting Media and Retail Investors

Broockman and Kalla (2025) both indicate that exposure to opposing political viewpoints can moderate extreme attitudes. The interaction encourages learning. If one seeks information and is conscious of the bias within it, it inspires itself to seek further view points to get away from that bias and get to the core information instead. Motivated reasoning often dampens this effect in polarized markets. In this matter, investors frequently reject information that conflicts with their prior beliefs. This reaction stems from what is known as cognitive dissonance where access to diverse news sources therefore does not guarantee reduced bias. This is why active critical engagement is required to overcome the tendency to filter out uncomfortable data.

The behavior of retail investors, particularly within online communities, provides a rich ground for studying sentiment, persuasion, and market dynamics. Alvarez et al. (2022), analyzing the r/WallStreetBets (WSB) community during the GameStop phenomenon, highlight how shared narratives (e.g., "David vs. Goliath," "stick it to the establishment") and common language can

foster a sense of unity and persuade individuals to undertake specific investment actions, such as buying and holding a stock against conventional financial logic. Factors such as fear of missing (FOMO) also play a significant role. This research highlights the need to differentiate between general support for a community or narrative and the specific intent to act (e.g., purchase shares), as persuasion involves convincing individuals to "buy-in" both ideologically and financially. Online communities operate as a potent force where social dynamics amplify sentiment and external media narratives shape the discourse. This combination translates into tangible market impact, meaning that any quantitative study of these interactions requires specialized datasets where models can capture both the expressed sentiment and the underlying intent.

AI-Driven Market Volatility Analysis

Abdollahi (2024) demonstrates how AI quantifies the influence of media sentiment on market volatility. The study employs advanced econometric models alongside NLP. These tools reveal the temporal and spatial dynamics of fluctuations driven by sentiment. This highlights a specific requirement. Predictive frameworks need to integrate political and emotional biases to capture the chaotic nature of stock market behavior.

Financial news spreads rapidly and acts as a catalyst for volatility. Predicting this movement presents a challenge. Gandhmal and Kumar (2019) review the computational techniques used to address it, such as AI and big data analytics. Salbrechter (2024) adds nuance to the discussion. Research regarding the reaction to overnight news found that sentiment has a clear impact on opening prices because the market absorbs these emotional signals early and processes them long before the opening bell actually rings. Instances of overreaction and underreaction at the market open are influenced by the relationship between returns from the previous day and overnight news sentiment so this leads to predictable return reversals during the trading day that follows. This specific pattern suggests that market inefficiency driven by sentiment contributes to volatility. Research concerning the Dhaka Stock Exchange demonstrates that machine learning techniques are actually better suited than traditional models for capturing nonlinear trends in markets with high volatility. These analyses move beyond simple correlations to model the

complex mechanisms where sentiment translates into market movements and volatility even when it is amplified by bias.

Implications for Predictive Models

Integrating political bias into financial sentiment analysis sharpens the accuracy of stock price prediction models because standard frameworks frequently miss the ideological factors that drive investor behavior. This oversight yields incomplete forecasts so quantifying political bias through NLP techniques creates a nuanced input for predictive algorithms. Hasani (2024) and Abdollahi (2024) validate this method by demonstrating that enriched inputs help models anticipate market reactions during periods of polarization or for stocks with distinct political sensitivity.

Research consistently demonstrates that sentiment analysis improves predictive accuracy. News sentiment predicts stock returns. This signal appears strongest in smaller stocks where information diffuses slowly. Salbrechter (2024) indicates that the timing of news dictates subsequent returns because of market inefficiencies. Bias detection refines these signals. Lim (2025) explores systems aware of emotion. That research uses models like FinBERT to guide traders. It implies that predictive models benefit from understanding psychological drivers. Models developed for volatile markets already show improved accuracy through machine learning which suggests that forecasting capabilities will likely improve even further when these systems integrate the political context that shapes sentiment alongside behavioral patterns.

Advancements in Sentiment Analysis Techniques

Recent studies highlight significant advancements in sentiment analysis, moving beyond simple lexicon-based methods towards sophisticated deep learning architectures. Domain-specific models like FinBERT, pre-trained on large financial corpora, demonstrate superior performance in capturing the nuances of financial language compared to general models or traditional lexicon methods like VADER (Hasani & Aung Zaw, 2024). FinBERT's specialization allows it to understand unique terminology and context within finance, leading to more reliable sentiment assessments. However, general-purpose Large Language Models (LLMs) like the GPT series

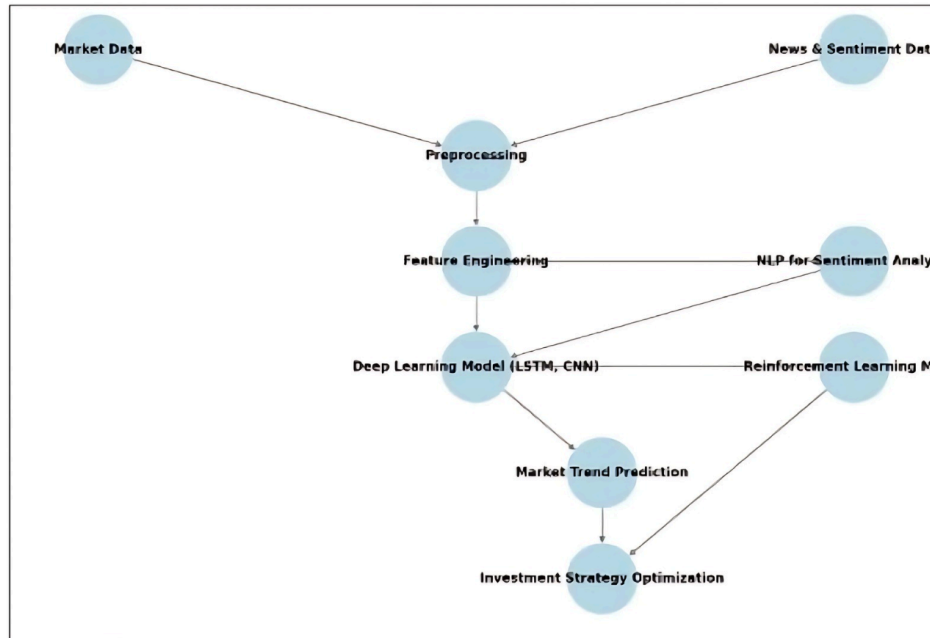
languages. Despite these strides, significant challenges persist regarding low-resource languages and cultural nuances (Renault, 2017; Pango, 2025). The research has consequently shifted toward finer granularity, utilizing Aspect Based Sentiment Analysis to isolate specific features and emotion detection to classify distinct feelings like anger or happiness rather than simple polarity. Accuracy remains elusive for sarcasm or irony. These nuances challenge even the most advanced models (Pango, 2025). Large Language Models like GPT 4 provide advanced context but introduce latency. High frequency settings struggle with the reproducibility of these tools. This study selects established lexicon architectures for that reason where TextBlob analyzes professional news and VADER processes social media to offer speed and transparency since generative models simply lack this deterministic nature.

Hybrid Models in Financial Market Forecasting

Hybrid forecasting models increasingly merge sentiment analysis with deep learning to handle time series data. Gu et al. (2024) exemplify this approach by pairing market psychology scores from FinBERT with LSTM networks that track historical price patterns. Islam and Monira (2025) note that this fusion proves particularly effective in volatile markets because it outperforms models capable of seeing only price data. Other architectures focus on robustness by combining ensemble methods like Random Forests with technical indicators such as MACD or RSI. Some studies even integrate Rough Sets with Artificial Neural Networks to achieve higher accuracy. Complexity is not a silver bullet because simpler models like Logistic Regression can still outperform dense architectures in specific tasks. Success requires correct tuning and high quality features from systems like GPT 4. These hybrid approaches ultimately rely on leverage since combining the contextual power of Transformers with the sequential modeling of LSTMs creates a toolset strong enough to handle modern market complexities. **Figure 3** presents a visual example of this hybrid framework.

Figure 3

AI-Powered Predictive Model for Market Trend Analysis



Note. This model outlines the process flow from data input (Market Data, News & Sentiment Data) through preprocessing, feature engineering, deep learning, and reinforcement learning to predict market trends and optimize investment strategy. From "Advanced computational forecasting techniques to strengthen risk prediction, pattern recognition, and compliance strategies", by N. T. Popoola & F. A. Bakare, 2024, *International Journal of Science and Research Archive*, 12(2), pp. 3033–3054 (<https://doi.org/10.30574/ijisra.2024.12.2.1412>). CC BY 4.0.

Market Dynamics and Retail Sentiment Integration

Understanding the interplay between market dynamics and retail investor sentiment is crucial because online platforms and social media now empower collective action while studies emphasize the significant role news sentiment plays in shaping these markets by directly influencing trading volumes and investor behavior (Lopez-Lira & Tang, 2023; Hasani & Aung Zaw, 2024).

Advanced language models accelerate information diffusion to reduce interpretation bias and bolster efficiency, but raw sentiment creates a contrasting dynamic where retail investors frequently ignite volatility. The r/WallStreetBets community provides a stark example of this since shared narratives on the forum drove investors to coordinate action against stocks like GameStop (Alvarez et al., 2022).

The sheer force of collective sentiment in this specific instance completely overwhelmed standard fundamental analysis which makes it incredibly difficult yet critical to distinguish between mere vocal support and actual trading intent. Traditional economic models really struggle with this dynamic because they operate on the presumption that rational actors always chase intrinsic value but the raw collective power of an online community simply invalidates that assumption. Hype cycles birth meme stocks where prices become totally detached from underlying fundamentals. When speculative bubbles form and inevitably crash they raise massive concerns about market stability. This highlights a potential necessity for regulatory measures that specifically target collective action driven by sentiment rather than financial rationale. These observations actually align perfectly with theories regarding noise traders and limited arbitrage. Traders driven by sentiment coordinate around specific narratives to exert massive price pressure even while fundamentals stay static. Professional arbitrageurs often hesitate to step in and correct these deviations because they fear the risk that the mispricing will intensify significantly before it finally resolves (Black, 1986; De Long et al., 1990).

Online communities simultaneously create herding dynamics. Visible participation acts as social proof. This feedback loop generates a cascade where users adopt positions or beliefs en masse (Banerjee, 1992; Bikhchandani et al., 1992). Such mechanisms validate the inclusion of retail sentiment and volume in predictive models for the short term. Narratives evolve online with speed. Sentiment shifts rapidly. Traditional analytical tools struggle to track these changes. The environment necessitates dynamic systems capable of monitoring data in real time to capture sudden shifts in market impact.

Real-time sentiment analysis, integrating structured data with sentiment indicators from news and social media, is vital for understanding these dynamics (Islam & Monira, 2025). Specific market contexts, like the Dhaka Stock Exchange (DSE), which is characterized by high volatility and significant retail investor participation, benefit greatly from ML models that can integrate sentiment. Furthermore, sentiment is not just a market phenomenon but also correlates with broader economic activity, with measures like the Michigan Consumer Sentiment Index (MCSI) predicting GDP trends (Kerssens, 2024). News-based sentiment indices, constructible in near real-time using NLP, show strong correlation with traditional surveys like MCSI and can

potentially capture turning points more quickly, offering valuable insights for both market participants and policymakers. The inefficiency in how quickly markets price in news sentiment, especially for smaller stocks, creates opportunities and challenges (Salbrechter, 2024). Capturing these complex dynamics requires integrating sentiment and behavioral patterns into market models. Political context also plays a key role. Relying on sentiment alone often misleads the analyst. The source dictates the validity of the signal. Algorithms must determine if a belief is genuine or the result of coordinated manipulation. Emotions might stem from hard facts or total misinformation. Real time sentiment analysis offers value but cannot function in isolation.

Anyone who studies the stock market knows that fundamental analysis provides a necessary counterweight since the broader social and political context remains essential to the stock movement. But here the core challenge involves building models that capture sentiment while distinguishing its origin because a reliable picture of market dynamics emerges only when algorithms separate organic belief from artificial noise.

This literature suggests a well developed structure for modern predictive modeling with three components where sentiment analysis captures the emotional tone of the market while political bias quantification isolates the systematic framing effects of information supply. This synthesis motivates the evaluation of the incremental predictive value of each component so that bias, sentiment, and retail noise compete against a baseline of technical efficiency.

Summary and Empirical Expectations

Synthesizing the research leads us to three specific expectations. Retail sentiment appears to fundamentally align with Noise Trader Theory since high volume sentiment quantified via VADER serves as a robust predictor for short term price direction. This specific factor holds the power to completely eclipse other market signals during viral events. The reputational prior hypothesis introduces a distinct role for political bias where the ideological orientation of publishers contributes incremental predictive utility. MBFC data captures this systematic information framing to ensure the element functions independently from mere sentiment polarity. Market regimes seem to control the way these features interact because retail noise tends to

affect the impact of subtle editorial bias when volatility is high. The signal from the publisher simply drowns in the massive volume of the crowd.

Methodology

This research implemented a reproducible machine learning pipeline to determine if media signals effectively improve stock direction forecasting for the next trading day. The analysis treats the prediction task as a binary classification problem because the target variable simply indicates whether the adjusted close price rises above the current level for a specific ticker. Three distinct data streams merge to form the modeling dataset where technical indicators provide the baseline from historical prices while article metadata supplies daily news sentiment and political bias. Reddit post volume and sentiment simultaneously proxy the behavior of the retail investor to capture the mood of the crowd. Scripted processing manages the entire workflow to ensure that every step remains traceable and replicable.

The complete source code for the dataset and modeling pipeline is open-source and available at: https://github.com/daniel-st3/Daniel_Rodriguez_MSc_Thesis_Final

Data Collection and Sources

For data collection three distinct data streams were found and aligned at the ticker-day level to construct the dataset. Stock market data (using the finance library) formed the structural backbone, where daily price observations defined the trading calendar for each ticker and served as the primary basis for computing technical indicators and prediction targets. This financial foundation was augmented by news articles (found using the GNews API connection) consolidated from raw files into a master table containing metadata such as titles, timestamps, and publisher names. To ensure data quality, a filtering process based on MBFC acted as a reputation proxy, retaining only outlets with higher factuality ratings while assigning standard ideological bias labels to the remaining publishers. To get retail trader discourse it was simultaneously captured through the Reddit API across various finance-related communities that were chosen by hand by applying ticker-specific matching patterns within the study's timeframe,

the system aggregated submission titles, scores, and comment counts to represent the sentiment and volume of the online crowd.

Data Governance and Ethical Considerations

The data collection process complies to the Terms of Service for the Reddit API and the Google News API to ensure full compliance throughout. User privacy was a priority so the project avoided collecting Personally Identifiable Information regarding Reddit users entirely. The analysis focused only on aggregate metadata like scores or timestamps instead of tracking individual histories. Finally, all news content came from legal public endpoints, the final dataset sits in local storage to match standard academic protocols for data protection.

Table 1

Final Modeling Sample Summary (after filtering)

Item	Value
Date range	2024-06-20 to 2025-06-30
Tickers included	10
Trading days (unique dates)	257
Ticker-days (rows)	2,570
Target class balance (Up / Down)	53.3% / 46.7%
Ticker-days with ≥ 1 news article	79.20%
Ticker-days with ≥ 1 Reddit post	52.20%
Total news articles	7,578
Total Reddit posts	3,147

The unit of analysis focuses on the ticker day to generate 2,570 observations across 10 tickers over the period from June 20, 2024 to June 30, 2025. The target variable tracks next day direction with a class balance of 53.3% up days versus 46.7% down days. News features cover 79.2% of these observations through 7,578 total articles while Reddit features appear in 52.2% of

the sample via 3,147 total posts because this distribution is highly relevant for interpreting the incremental contribution of sentiment derived signals.

Feature Engineering

All engineered features (detailed in the **Data Dictionary in Appendix A**) were computed at a daily ticker level and later merged onto trading dates.

Technical indicators (from prices)

For each ticker the analysis computed indicators using adjusted close prices where the RSI utilized a 14 day window based on rolling averages of gains and losses. The MACD was derived simultaneously as the difference between the 12 period and 26 period exponential moving averages with its signal line calculated as a 9 period EMA of the resulting series. Rows lacking sufficient lookback history for the RSI were excluded to ensure that no missing values entered the modeling dataset.

Target variable (next-day direction)

The target variable (target_up) is only a binary marker, meaning, it gets a 1 if the next day's adjusted close beats the current price, and a 0 if it doesn't. So, it is compulsory to know because this logic relies on knowing what happens the following day, the very last observation for every ticker had to be removed. This of course because there just wasn't a "next" price to compare against for those specific rows, so they lacked a future label and were dropped from the set.

News sentiment and political-bias features (ticker-day aggregation)

Once publishers were filtered to the retained factuality categories, article level sentiment was computed using the TextBlob library because its lexicon based approach efficiently processes large volumes of unstructured text. Although Transformer based models like FinBERT provide superior contextual awareness, TextBlob was selected for this module to minimize computational latency and effectively mimic the strict constraints of a high frequency production environment.

Publisher bias labels were mapped onto a numeric scale to represent ideological direction using a five point encoding that spans from Left at negative two up to Right at positive two. This distinction is necessary because MBFC provides separate dimensions for ideological political bias versus factuality or reliability. The analysis defines political bias strictly as the ideological dimension so labels like Questionable are excluded since they belong to the factuality category rather than representing a specific political leaning. These article level fields were subsequently aggregated into daily ticker features that quantify the mean article sentiment and the total count of retained articles per day. The final dataset also includes the mean bias score to capture the direction of ideological leaning and the standard deviation of that score to measure ideological dispersion.

The raw dataset contains a distinct blend of bias labels such as Left alongside factuality markers like Questionable. This study categorizes the Questionable label strictly as an indicator of factuality rather than placing it on the ideological bias spectrum. These specific sources remain within the dataset to capture high velocity retail narratives. The methodology avoids coercing them into a political leaning category like Left or Right. Their potential impact is instead captured partially through the dispersion metric.

More specific separation of ideological bias from factuality (e.g., robustness checks excluding low-reliability outlets) is left for future work since this inclusion is deliberate, as 'Questionable' sources may contribute to the viral retail narratives under investigation, also, it is important to note, when no articles were available for a ticker-day, news features were imputed with 0 (neutral), so this method assumes that 'no news is good news' (stationarity), and it was chosen to be this way to prevent data leakage associated with forward-filling in a volatile time-series context.

Deciding to measure political bias at the publisher level instead of looking at individual articles was a major choice in how this study was designed. This approach treats bias as something more than just the words on a page: it is a reputational prior that is already baked into how the market processes information.

In fast paced trading environments, both people and algorithms react to the brand identity of a news source almost instantly. For example, seeing a headline from the Wall Street Journal versus the Guardian triggers a reaction well before the actual meaning of the article is fully processed. The source itself becomes a shortcut that the market uses to anticipate the slant of the news.

Therefore, the publisher's established ideological stance functions as a heuristic or "prior" that frames investor interpretation immediately upon publication. By modeling this reputational attribute directly, the feature set captures the market's behavioral reaction to the source itself, distinct from the granular sentiment of the text. While publisher-level labels inherently contain measurement noise compared to article-specific stance detection, this approach tests the robustness of the "brand signal" itself. The consistent predictive lift observed across multiple folds suggests that even this high-level proxy captures a stable, predictively useful component of market information processing, essentially functioning as a noise-reduced prior for algorithm inputs. It is important to note that these labels act as a third-party heuristic rather than an objective ground truth; they are employed here to model the market's reaction to a perceived 'reputational prior' rather than to validate the publishers' actual editorial neutrality. While MBFC is a third-party aggregator, it is widely utilized in computational social science research (e.g., Baly et al., 2018) as a proxy for domain-level ideological reliability, offering a standardized heuristic for publisher reputation.

MBFC labels: political bias vs factuality

MBFC attribute	Categories	Variables in thesis	Application in analysis
Political bias (ideological leaning)	Left, Left Center, Least Biased, Right Center, Right	bias_score (e.g., -2 to +2), disp_bias	This metric is the core ideological signal for forecasting it also enables the testing of interaction effects
Factuality / reliability	Very High, High, Mostly Factual, Mixed, Low, Very Low, Questionable	factuality_label and/or is_questionable	Utilized strictly as a binary marker (is_questionable) to track source reliability exclusion of these outlets remains a step for future sensitivity analysis

The MBFC framework provides distinct categories for ideological bias and factual reporting reliability. Within this thesis political bias refers exclusively to the ideological bias dimension which is encoded on a scale from -2 to +2. The Questionable label belongs to the factuality dimension instead and is therefore not interpreted as a specific political leaning. When relevant factuality is managed by using a flag to identify questionable sources or by suggesting that future work can test robustness by excluding these sources entirely.

Retail sentiment features

To get the Reddit data ready, the submission titles and the actual body text were joined together into a single string for each post. Sentiment was then pulled out using the VADER compound polarity score and grouped into daily stats for every ticker. One of the main metrics is the retail sentiment, which is just the average VADER score for a ticker on a given day. Another is the post count, which tracks the total number of submissions daily to see how much noise or volume there is in the conversation.

Since the final dataset only focuses on trading days, these daily features from Reddit had to be mapped forward to the next available market session. This means that talk from the weekends or holidays still contributes to the model by being assigned to the following Monday. If several days worth of Reddit chatter mapped to the same trading day, the sentiment scores were averaged and the total number of posts were summed up. Any gaps left over after the merge were filled with zeros to keep the feature matrix consistent.

The final modeling table was saved as `3_final_modeling_dataset.csv` with one row per ticker and trading day, containing technical indicators, news-based features, retail features, and the target label. So while the main model used strict date-blocked cross-validation the interaction effect analysis employed a randomized pooled cross-validation to maximize signal detection across tickers meaning these specific interaction findings should be interpreted as descriptive of the dataset's internal structure rather than a strictly simulated out-of-sample trading forecast.

Modeling and Validation

Feature sets (ablation design)

To really pin down how much political bias and social media chatter actually add to the models, four different feature sets were tested. This approach shows exactly where the predictive power is coming from, moving step-by-step from basic indicators to the more complex behavioral signals.

- **Baseline:** This is the core setup using only standard technical indicators like RSI, MACD, and volume to establish a starting point using traditional technical analysis.
- **Bias Only:** This model focuses strictly on the use of political slant of the news coverage to see if that metadata carries weight on its own.
- **Retail Only:** This set looks purely at what is happening on Reddit, using sentiment scores and post volume to represent the "crowd" effect.
- **Enhanced:** This final set combines everything, the technicals, the news bias, and the retail sentiment to see how they perform when they all work together.

Models

To have a well structured answer three classifiers were evaluated, Logistic Regression, Random Forest and Gradient Boosting. Logistic Regression was trained within a pipeline with feature standardization (StandardScaler) to ensure coefficient comparability across predictors. It is also important to note class imbalance was handled through algorithm-level weighting (Logistic Regression with `class_weight="balanced"` and Random Forest with `class_weight="balanced_subsample"`), rather than through synthetic oversampling.

Time-series validation

Model evaluation used date-based time series cross-validation with five sequential splits. Splits were created on the ordered set of unique trading dates, and for each fold all tickers observed on the training dates were used to train the model, while all tickers observed on the held-out dates were used for testing. This design preserved temporal order and avoided look-ahead bias (see

code implementation in Appendix B.3). Due to the limited one-year sample window, the statistical significance tests utilized a 5-fold time-series split. While this provides an indication of stability across different market regimes, the p-values should be interpreted with caution given the limited degrees of freedom.

In addition to learned models, several naive benchmarks were also computed using the same time splits: always predicting DOWN, always predicting UP, a global training-set majority-class rule, a per-ticker training-set majority-class rule, and a persistence baseline using the previous day's realized direction for the same ticker. All of this to have a more general view.

Evaluation

Model performance was evaluated using a set of classification metrics that capture both overall correctness and class-sensitive behavior that are used in these sort of investigations. Accuracy, precision, recall, F1 score for the UP class, macro-averaged F1, balanced accuracy, ROC AUC (when defined), and precision–recall AUC (average precision, when defined). So results were reported at the fold level and aggregated across folds using mean performance, with variability summarized via the standard deviation across folds. To assess whether feature augmentation produced consistent improvements, statistical comparisons between feature sets were conducted using bootstrap resampling of fold-level metric differences, producing confidence intervals for the mean improvement and a two-sided bootstrap p-value for each model and metric. Outputs included fold-level metrics, aggregated comparison tables, and out-of-fold prediction files to support later error analysis and visualization.

Mathematical Formulation of Classifiers

To ensure further reproducibility and theoretical transparency, this section will detail the specific mathematical objective functions optimized by the three machine learning architectures employed in this study, it is critical to define these functions to understand how each model treats the noise inherent in financial text data.

Logistic Regression

Logistic Regression serves as the linear baseline for this study. It models the probability $P(Y=1|X)$ that the target variable Y (next day price increase) belongs to class 1 given the feature vector X . The model assumes a linear relationship between event log odds and predictors. The mathematical formulation is as follows:

$$\ln\left(\frac{P(Y=1|X)}{(1 - P(Y=1|X))}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

The probability is estimated using the sigmoid function $\sigma(z)$ which maps any real valued number into a value between 0 and 1:

$$\sigma(z) = \frac{1}{(1 + e^{-z})}$$

In this equation z represents the linear combination of weights and features $\beta^T X$. The model parameters β are estimated by maximizing the log likelihood function $L(\beta)$. This function measures how well the model predicts the observed data:

$$L(\beta) = \sum_{i=1}^n [y_i \ln(\sigma(\beta^T x_i)) + (1 - y_i) \ln(1 - \sigma(\beta^T x_i))]$$

To prevent overfitting given the high dimensional noise in financial text data, L2 regularization or Ridge regularization was applied. This modifies the objective function to minimize the cost function $J(\beta)$ rather than maximizing likelihood directly. The regularization term penalizes extreme parameter weights:

$$J(\beta) = -L(\beta) + \lambda \sum_{j=1}^k \beta_j^2$$

In this formula λ represents the regularization parameter. This term penalizes large coefficients and effectively shrinks the influence of collinear features. This is a critical step given the correlation between technical indicators like MACD and MACD Signal ($r=0.95$) in the dataset.

Random Forest Classifier

The Random Forest model (RF) constructs an ensemble of B decision trees $\{T_1, T_2, \dots, T_B\}$. Each tree is trained on a bootstrap sample of the dataset using a method known as bagging. For a given node t in a tree, the split is chosen to maximize the decrease in impurity. This study utilizes Gini Impurity as the splitting criterion. For a node t with class probabilities p_i the Gini impurity $I_G(t)$ is defined as:

$$I_G(t) = 1 - \sum_{j=1}^C p_j^2$$

In this context $C=2$ for Binary Classification (Up vs Down). The "information gain" from a split θ that partitions node t into left (t_L) and right (t_R) is calculated as follows:

$$\Delta I(\theta, t) = I_G(t) - \left(\frac{N_L}{N} I_G(t_L) + \frac{N_R}{N} I_G(t_R) \right)$$

In this equation N represents the total samples in the parent node while N_L and N_R represent the samples in the child nodes. The Random Forest prediction $\hat{Y}$ for a new instance is determined by majority voting across all B trees:

$$\hat{Y} = \text{mode} \{ T_1(x), T_2(x), \dots, T_B(x) \}$$

This ensemble approach reduces the variance associated with individual decision trees. It makes the model robust to outliers in the social media sentiment data which often contains extreme values.

Gradient Boosting Classifier

Unlike Random Forest which builds independent trees, Gradient Boosting constructs an additive model in a forward stage wise fashion. At step m , a new weak learner (tree) $h_m(x)$ is fit to the pseudo residuals of the previous model $F_{m-1}(x)$. The objective is to minimize a differentiable loss function $L(y, F(x))$. For binary classification the model optimizes the Log Loss or Deviance:

$$L(y, F(x)) = - (y \ln(p) + (1 - y) \ln(1 - p))$$

In this formula $p = \sigma(F(x))$. The update rule for the model at step m introduces a shrinkage parameter to control overfitting:

$$F_m(x) = F_{m-1}(x) + v * h_m(x)$$

Here v is the learning rate where $0 < v \leq 1$. This shrinkage parameter v scales the contribution of each tree and prevents the model from fitting the training noise too quickly. In this study Gradient Boosting proved prone to overfitting the majority class. Its high Recall indicated this but low AUC of 0.506. This suggests that the boosting process aggressively minimized loss on the dominant "Up" days at the expense of discriminatory power on "Down" days.

Empirical Results

This chapter presents the empirical results in two stages: (i) descriptive evidence about the constructed news corpus and the derived textual signals, and (ii) diagnostic evidence about the daily feature space used for next-day stock direction forecasting.

Note on Terminology: To keep the terminology clear it is important to note that the Baseline Feature Set in this analysis refers specifically to the model trained only on technical indicators like RSI and MACD along with volume. In contrast the term Naive Baselines describes non predictive strategies such as Always Up or Persistence which are used purely for benchmarking the actual performance of the model.

Descriptive Analysis of the Data

The final collection of news data includes 8,830 articles that were put out between May 31, 2024, and June 1, 2025. These stories came from 229 different publishers. To make sure the analysis stays easy to follow, the factuality scores from MBFC are reported separately from the actual political bias labels.

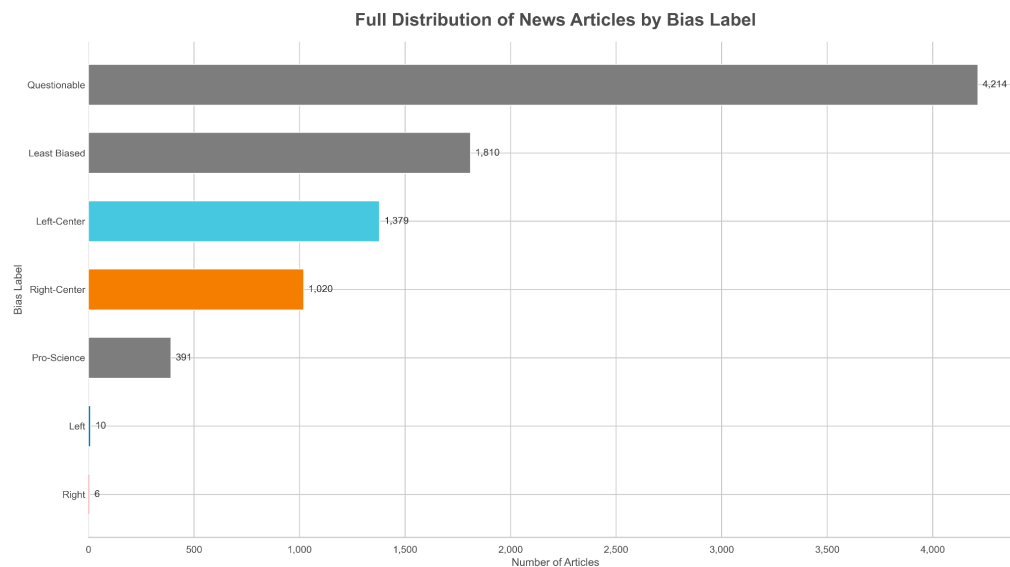
A huge portion of this data comes from sources that are flagged for reliability problems. The 'Questionable' group accounts for 4,214 articles: roughly 47.7% of the entire set. This really highlights the amount of high-noise information that retail traders are exposed to every day. These specific articles are marked with the `is_questionable` flag and aren't assigned a spot on the political spectrum when looking at the bias-specific results.

The remaining 4,616 articles were mapped onto a 5-point ideological scale. The 'Center' group, which pulls together the 'Least Biased' and 'Pro-Science' labels, is the most common with 2,201 articles. Following that are 'Left-Center' at 1,379 and 'Right-Center' at 1,020. Extreme views are actually quite rare in this specific high-frequency sample: 'Left' only shows up 10 times and 'Right' only 6. This distribution shows that the real ideological variety in the data is mostly a split between moderate-left and moderate-right framing.

The counts for these labels match the article totals exactly after cleaning up the extra spaces in the raw bias_label field. This confirms the cleaning process fixed the formatting without messing with the actual numbers. It is also worth noting that the bias_label itself is based on the categories assigned to the publisher and then applied to every article they produce.

Figure 4

Distribution of News Articles by Political Bias Category



Note. Author's own visualization of MBFC label coverage in the news corpus (n = 8,830).

Sentiment Differences Across Political Bias Groups

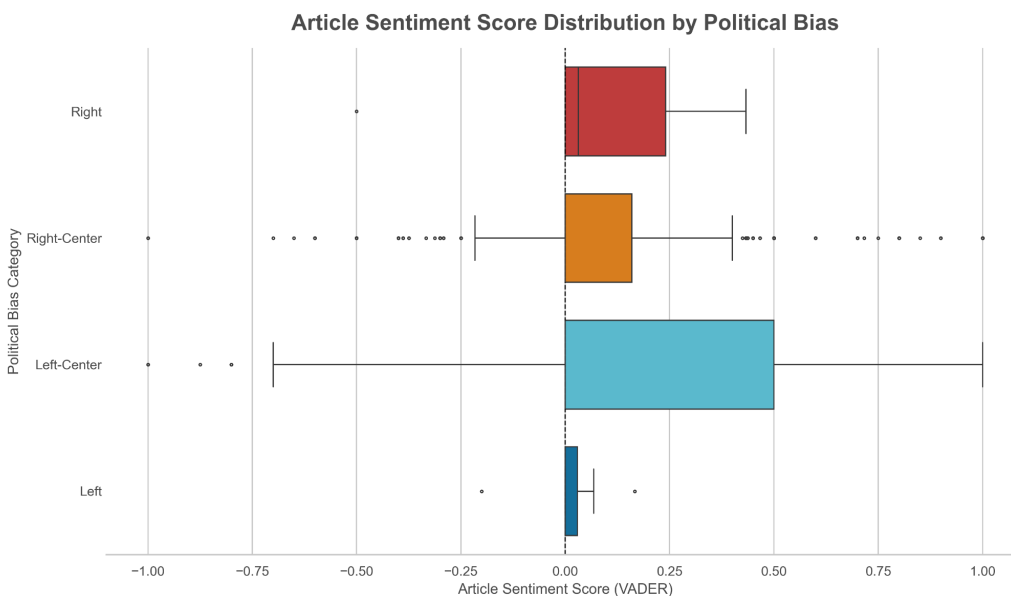
Sentiment comparisons are conducted on the four political bias groups displayed in the sentiment distribution figure: Left (n=10), Left-Center (n=1,379), Right-Center (n=1,020), and Right (n=6). Running a Kruskal Wallis test on article level sentiment across these groups highlights distinct differences in distribution ($H = 60.265$, $p = 5.161e-13$). But here is the nuance. If looking strictly at the summaries, the medians actually hover quite close to neutral for most categories. Left, Left Center, and Right Center all sit flat at 0.000. The Right category is the outlier here, showing a slight shift with a median of 0.031.

If you look closely at the upper end of the spread, a distinct split emerges that standard averages tend to obscure, while the medians sit close together, the third quartile tells a different story in here Left-Center sources climb to a score of 0.500, whereas Right-Center outlets sit significantly lower at 0.160. This gap feels very structural since it suggests that positive sentiment doesn't just spread evenly across the media landscape, it clusters. Specifically, this optimism bundles up within Left-Center narratives in a way that isn't mirrored on the right side of the spectrum. This directly means if focus is centered in central tendencies the sentiment is missed entirely since the real signal isn't sitting safely in the middle of the distribution it is in its tails.

The far Left and far Right groups are different. Sample sizes here are very small. This makes it hard to prove anything solid about them. These points are just descriptive. Most of the useful evidence comes from the center categories. There is a sentiment gap in mainstream news. No single pattern covers the whole political spectrum.

Figure 5

Article Sentiment Score Distribution by Political Bias



Note.

Author's own visualization of TextBlob news sentiment by bias group.

Temporal Patterns in News Supply by Ideological Category

Weekly news volume is grouped into Sunday-ending bins. This uses the `resample('W')` operation on timestamps. The time series covers 53 weeks in total. It starts June 2, 2024, and runs until June 1, 2025. This aggregation is just for description. It does not change the actual article-level data.

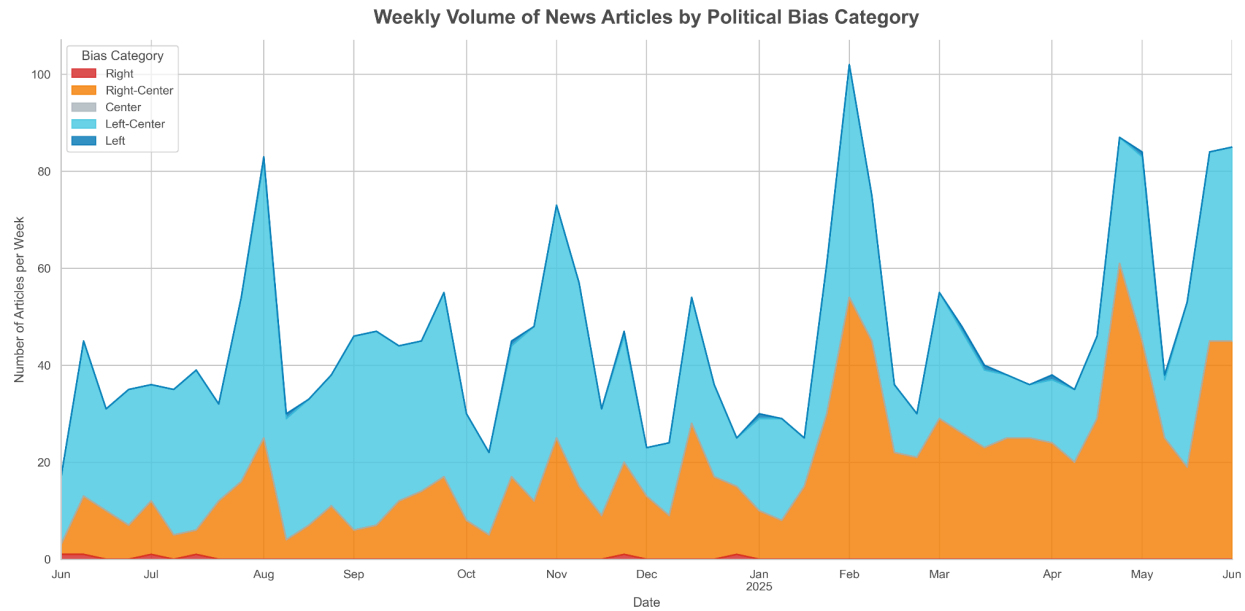
The label "Least Biased" is switched to "Center" for the figures. Underlying source classifications stay the same. News output is not a smooth line. Activity shows up in concentrated bursts. These bursts define the shape of the whole series.

The supply spike occurred between January 27 and February 2, 2025, with a total volume of 194 articles. However, it should be noted that the surge did not spread evenly across the political spectrum. Rather, it skewed heavily toward the middle. Center outlets produced 79 pieces during this period, while Right-Center sources contributed an additional 61 articles. Left-Center trailed behind with just 54. So basically, moderate sources drove the entire growth cycle. Market behavior during that window seemed to track the volume pretty closely. That concentration in center-leaning outlets likely worked as a narrative anchor, something for everyone else to reference. High article counts usually line up with retail attention, and the data backs this up. No other segment even came close to that kind of output.

What stands out here is that the fringes, both Far Left and Far Right were completely silent during this peak. It seems that when attention is highest, it is the mainstream, center-leaning publishers doing the heavy lifting, while the ideological extremes effectively sit on the sidelines.

Figure 6

Weekly Volume of News Articles by Political Bias Category



Note. Author's own visualization of weekly article volume by bias group.

Daily Feature Structure and Multicollinearity Diagnostics

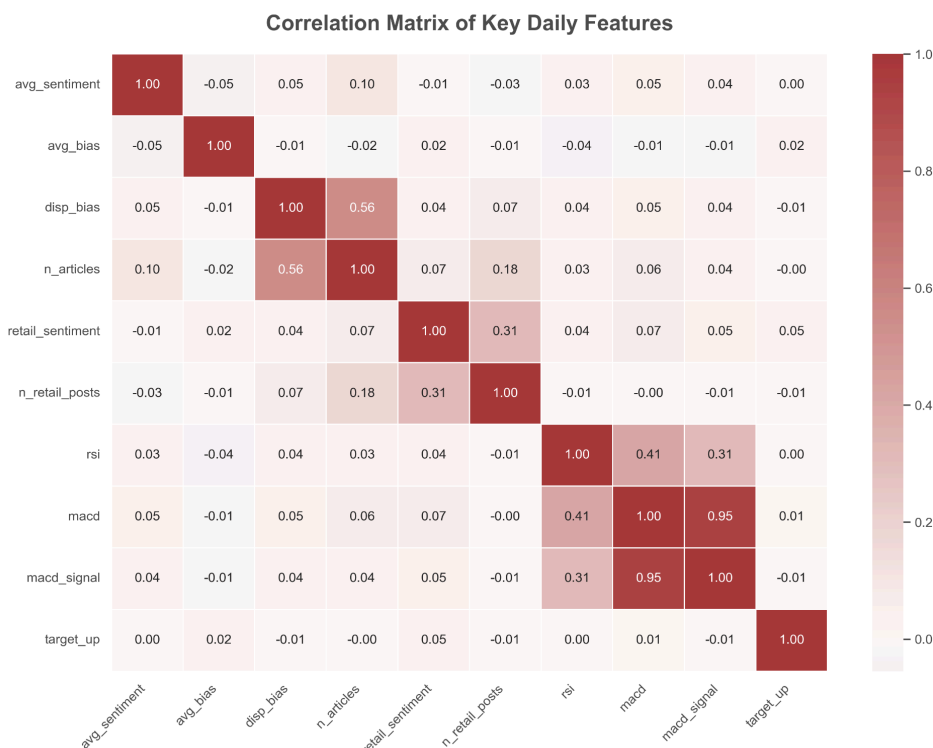
Correlation diagnostics are run on the final modeling dataset. This includes features like `avg_bias`, `retail_sentiment`, and `macd`. The strongest link is between `macd` and `macd_signal` with an r of 0.948. This makes sense given how the signal is built. Other notable correlations show up too. `disp_bias` links with `n_articles` at 0.556. `retail_sentiment` and `n_retail_posts` show a correlation of 0.311.

When we examine the relationship between technical indicators and textual variables, the correlations turn out to be quite small. In fact, the highest absolute link across the blocks is only 0.065. You see the same weakness when looking at links with the binary target: `retail_sentiment` and `target_up` show the strongest connection, yet even that sits at a mere 0.046.

It is important to keep in mind that these results essentially just show linear pairwise dependence. They don't act as the final word on multivariate importance. For reference, all these numbers are pulled from a total of 2,570 ticker-day observations.

Figure 7

Feature Correlation Heatmap



Note. Author's own visualization of Pearson correlations computed on the final modeling dataset ($n = 2,570$ ticker-days).

Predictive modeling setup and evaluated feature sets

Model results sit in one consolidated file. It groups outcomes by family and feature set. This layout makes comparison simple. Three specific families are tested: Logistic Regression, Random Forest, and Gradient Boosting. Each one goes through four setups. These are baseline, bias-only, retail-only, and enhanced.

The evaluation uses standard classification metrics. Accuracy, precision, recall, F1, and AUC. Since the dataset is imbalanced, balanced accuracy and PR AUC are probably the best indicators. Results are aggregated across folds instead of just using a single pooled score. This lets us check

for stability across different subsets of the data. The dispersion metrics, like `f1_std` and `f1_macro_std`, give us insight into the variance. These metrics show how much performance fluctuates between different market periods. This variability matters because it tells us whether the model stays consistent or if it's sensitive to changing market conditions. A model might look good on average but still swing wildly depending on the time period, which is something you need to know before deploying it in practice.

Statistical tests remain separate from the main metrics. This method uses bootstrap deltas compared against a reference. While using five cross-validation folds limits statistical power, the choice ensures patterns remain consistent across different data slices. This prevents the results from being a fluke of one specific partition.

Table 2

Model Comparison Results

Model	Feature Set	Accuracy	Precision	Recall	F1 Score	AUC	Balanced Accuracy	F1 Macro	PR AUC	F1 Std	F1 Macro Std
Logistic Regression	retail_only	0.516	0.553	0.554	0.543	0.523	0.521	0.511	0.565	0.037	0.011
Logistic Regression	enhanced	0.515	0.550	0.578	0.554	0.522	0.520	0.509	0.564	0.028	0.015
Random Forest	bias_only	0.511	0.538	0.619	0.568	0.515	0.510	0.499	0.541	0.046	0.021
Random Forest	enhanced	0.512	0.539	0.649	0.582	0.514	0.510	0.496	0.544	0.025	0.014
Random Forest	retail_only	0.505	0.533	0.634	0.572	0.508	0.504	0.490	0.539	0.035	0.009
Gradient Boosting	enhanced	0.510	0.534	0.688	0.596	0.506	0.503	0.483	0.545	0.047	0.035
Gradient Boosting	bias_only	0.510	0.534	0.685	0.593	0.506	0.505	0.484	0.540	0.050	0.028
Logistic Regression	baseline	0.499	0.534	0.542	0.521	0.502	0.505	0.486	0.542	0.076	0.030
Random Forest	baseline	0.505	0.534	0.598	0.558	0.501	0.504	0.494	0.526	0.043	0.014
Logistic Regression	bias_only	0.487	0.522	0.555	0.525	0.497	0.492	0.476	0.542	0.064	0.032
Gradient Boosting	baseline	0.506	0.530	0.675	0.587	0.497	0.499	0.479	0.531	0.062	0.031
Gradient Boosting	retail_only	0.502	0.529	0.657	0.581	0.492	0.497	0.481	0.532	0.039	0.023

Table 3*Significance Tests Metrics*

Model	Metric	Comparison	Mean Delta	CI Low	CI High	Bootstrap p	N Folds
Gradient Boosting	auc	bias_only - baseline	0.009	0.001	0.015	0.03	5
Gradient Boosting	auc	enhanced - baseline	0.009	-0.006	0.021	0.196	5
Gradient Boosting	auc	enhanced - bias_only	0.001	-0.011	0.007	0.528	5
Gradient Boosting	auc	enhanced - retail_only	0.014	-0.008	0.035	0.187	5
Gradient Boosting	auc	retail_only - baseline	-0.005	-0.025	0.019	0.61	5
Logistic Regression	auc	bias_only - baseline	-0.005	-0.015	0.01	0.015	5
Logistic Regression	auc	enhanced - baseline	0.02	0.013	0.026	0.005	5
Logistic Regression	auc	enhanced - bias_only	0.025	0.015	0.037	0.001	5
Logistic Regression	auc	enhanced - retail_only	-0.001	-0.004	0.002	0.569	5
Logistic Regression	auc	retail_only - baseline	0.021	0.014	0.028	0.002	5
Random Forest	auc	bias_only - baseline	0.014	0.005	0.022	0.015	5
Random Forest	auc	enhanced - baseline	0.013	0.013	0.033	0.077	5
Random Forest	auc	enhanced - bias_only	0.009	-0.004	0.022	0.976	5
Random Forest	auc	enhanced - retail_only	0.006	0.007	0.027	0.236	5
Random Forest	auc	retail_only - baseline	0.007	0.003	0.011	0.380	5
Gradient Boosting	accuracy	bias_only - baseline	0.008	-0.003	0.018	0.076	5
Gradient Boosting	accuracy	enhanced - baseline	0.007	-0.009	0.021	0.208	5
Gradient Boosting	accuracy	enhanced - bias_only	-0.001	-0.015	0.012	0.535	5
Gradient Boosting	accuracy	enhanced - retail_only	0.01	-0.012	0.032	0.176	5
Gradient Boosting	accuracy	retail_only - baseline	-0.003	-0.026	0.017	0.603	5

Main predictive performance results

Across the 12 configurations in model_comparison_results.csv, the observed metric ranges are:

Accuracy	F1	AUC	PR_AUC
0.487 (Min)	0.521 (Min)	0.492 (Min)	0.526 (Min)
0.516 (Max)	0.596 (Max)	0.523 (Max)	0.565 (Max)

The best overall AUC was achieved by the Logistic Regression retail_only model (0.523431), while the Logistic Regression retail_only model also achieved the highest PR-AUC (PR-AUC = 0.565).

Within each model family, the best-performing feature set depends on the metric:

Model Family	Best AUC Metric	AUC Value	Best F1 Metric	F1 Value
Logistic Regression	retail_only	0.523	enhanced	0.554
Random Forest	bias_only	0.515	enhanced	0.582
Gradient Boosting	enhanced	0.506	enhanced	0.596

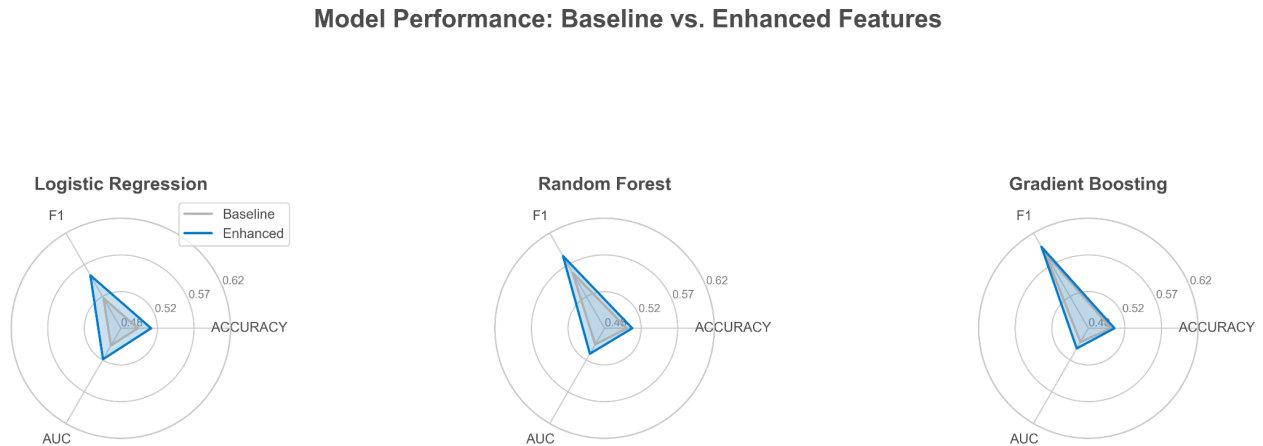
To quantify the effect of adding the full feature set, enhanced – baseline deltas were computed within each model family:

Model Family	Δ accuracy (Enhanced - Baseline)	Δ f1 (Enhanced - Baseline)	Δ auc (Enhanced - Baseline)	Δ pr_auc (Enhanced - Baseline)
Logistic Regression	0.017	0.033	0.020	0.022
Random Forest	0.007	0.024	0.013	0.018
Gradient Boosting	0.004	0.009	0.009	0.014

Overall, the highest discrimination metrics were achieved by the Logistic Regression model, specifically in the Retail-Only configuration (AUC = 0.523) and the Enhanced configuration (AUC = 0.522). While the Gradient Boosting model achieved the highest Recall (0.688) and F1-score (0.596) in the Enhanced setup, its lower AUC (0.506) indicates a tendency to over-predict the majority class ("UP") rather than true discrimination. This suggests that in a high-noise environment, the linear model (Logistic Regression) proved more robust at isolating the subtle signal provided by the textual features than the non-linear ensemble methods.

Figure 8

Model Performance Radar



Note. Author's own visualization based on model_comparison_results.csv.

Statistical significance of differences (bootstrap)

Bootstrap tests evaluate whether the enhanced feature set improves AUC beyond the baseline under the null hypothesis of zero performance difference ($H_0: \Delta AUC = 0$), with statistical significance determined at the $p < 0.05$ level. As detailed in **Table 3**, statistical significance was assessed using a non-parametric bootstrap test on the fold-level AUC differences (n=5 splits, 5000 iterations). This approach avoids assumptions of normality and rigorously tests whether the observed lift is consistent across different time periods. Paired bootstrap resampling indicates that the Retail-Only configuration improves AUC vs baseline (Mean Delta = +0.021, $p = 0.001$), and the Enhanced model also improves AUC vs baseline (Mean Delta = +0.020, $p = 0.005$). Random Forest shows a meaningful improvement in AUC. This occurs when the bias-only specification is compared with the baseline. The mean delta is +0.014 with a p-value of 0.015. That improvement does not carry over to Gradient Boosting. For that model, the difference between the enhanced version and the baseline is not statistically significant ($p = 0.196$). The contrast between these two results is informative. It suggests that simpler decision structures are better suited to extracting weak directional signals. More complex non-linear models may be

fitting noise rather than signal in this setting. It is also important to keep the sample size in mind. Granted, working with just five cross-validation folds doesn't exactly provide massive statistical power. Because of that hit, we have to be pretty cautious about how we interpret the results, it is crucial not to fall into the trap of overstating what is effectively a limited sample.

But looking at the bootstrap analysis, the model seems stable. Inside this window, anyway. Does that mean it works in every market regime? No. Definitely not. It just suggests the signal is real for this specific time. It's consistent here. It's unlikely to be a random fluke from one bad data split.

Benchmarking versus naive baselines

The naive benchmark results live in the file `naive_baselines_results.csv`. Two simple reference strategies are used here. Comparisons are made against the strongest model found in the main results. That model is Logistic Regression with the retail-only feature set. The benchmarks are the `always_up` rule and the `persistence_prev_day_same_ticker` strategy.

The `always_up` baseline puts up some confusing numbers. The F1 score sits high at 0.693256, the F1 Macro score is 0.346628. The balanced accuracy sits at 0.500000, matched perfectly by the AUC and lastly the PR AUC lands at 0.531905.

The mechanism here is obvious: the approach is simply forcing a prediction of the majority class every single time. That explains why the single-class F1 looks passable at first glance. But make no mistake, the model possesses zero discriminatory power. The 0.500000 scores for AUC and balanced accuracy allow for no other interpretation; the baseline is statistically indistinguishable from random chance. It's essentially a coin toss.

The persistence strategy tries to strike some balance, but the numbers still fall flat. F1 comes in at 0.523730, with F1 Macro at 0.490617. Balanced accuracy doesn't improve things its still stuck at 0.490706. AUC and PR AUC remain completely flat at 0.500000 and 0.531905. So while this baseline avoids the distortion seen in the always up strategy, it generates no lift whatsoever. Performance? Essentially another coin flip.

Zeroing in on the top performing setup, Logistic Regression with retail only features, shows it posting an F1 of 0.543440 and a Macro F1 of 0.511133. Balanced accuracy climbs to 0.521250 while AUC and PR AUC land at 0.523431 and 0.564763 respectively. While these gains might look modest at first glance, the crucial takeaway is that the lift is consistent across the metrics that actually handle class balance. That is a far better signal than just staring at raw accuracy.

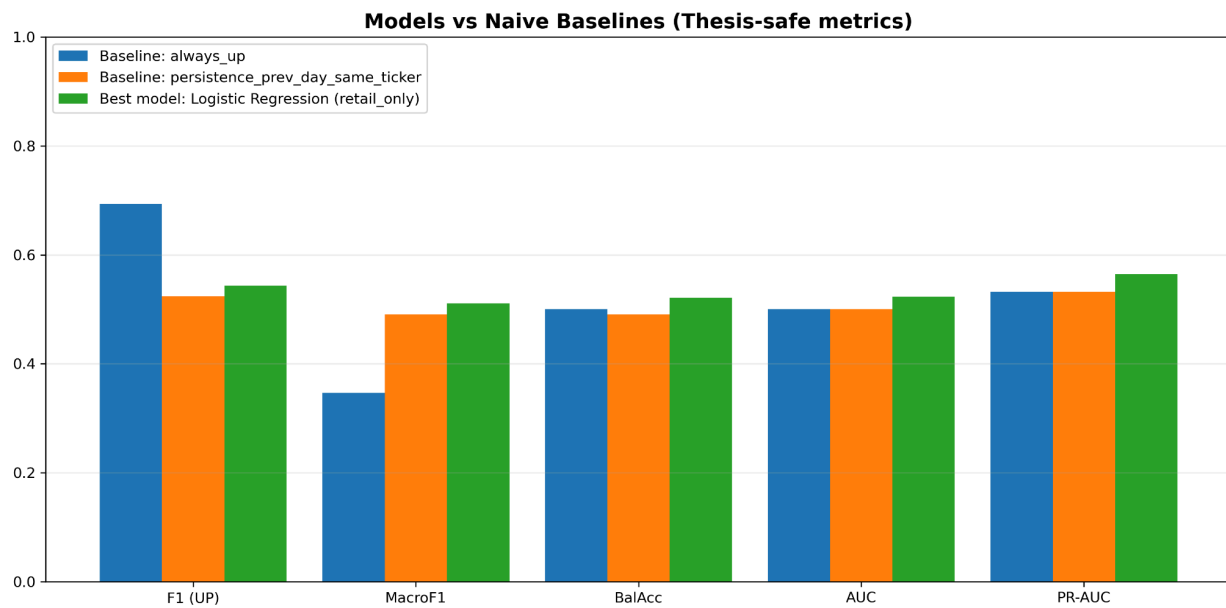
Stacking this against the always up strategy shows a drop in raw F1. The dip is - 0.149816. However the trade off is clear. F1 Macro jumps by a massive +0.164505. Clear gains appear elsewhere. Balanced accuracy rises +0.021250. AUC adds +0.023431. PR AUC climbs +0.032858. The story is similar against persistence, where F1 edges up by +0.019710, with F1 Macro and balanced accuracy following suit.

The retail only Logistic Regression model takes the win where it counts. Discrimination. It consistently outperforms both naive strategies on AUC. PR AUC and balanced accuracy follow suit.

By predicting Up every single time, the F1 score is artificially inflated. Of course, the figure might look impressive on a summary table, but in reality it is deceptive. The AUC remains stuck at 0.500000, confirming that the model has absolutely zero discriminatory power. It's a classic data trap. It has to be remembered that in imbalanced datasets, a high metric often signals noise rather than any genuine predictive skill. To look at a broader set of criteria to separate the signal from the class prevalence is a must. Otherwise, the performance metrics aren't showing capability, they might just be reflecting the dominant distribution.

Table 4*Naive Baselines Results*

Baseline	Accuracy	Precision	Recall	F1 Score	F1 Macro	Balanced Accuracy	AUC	PR AUC
always_down	0.468	0	0	0	0.318	0.5	0.5	0.532
always_up	0.532	0.532	1	0.693	0.347	0.5	0.5	0.532
majority_class_train_global	0.532	0.532	1	0.693	0.347	0.5	0.5	0.532
majority_class_train_per_ticker	0.518	0.53	0.917	0.666	0.381	0.496	0.484	0.525
persistence_prev_day_same_ticker	0.498	0.523	0.525	0.524	0.491	0.491	0.5	0.532

Figure 9*Models vs Baseline*

Note. Author's own comparison of best model vs. naive baselines.

Feature importance and coefficient-based interpretation

Feature coefficients are reported in `4_logreg_coefficients_enhanced.csv` for eight inputs:

- `macd`: 0.224
- `macd_signal`: -0.221
- `retail_sentiment`: 0.108
- `avg_bias`: 0.036
- `n_retail_posts`: -0.053
- `rsi`: -0.023
- `disp_bias`: -0.011
- `avg_sentiment`: 0.002

Feature coefficients from the standardized Logistic Regression model (Enhanced) reveal a clear hierarchy of signal strength. As expected, technical indicators dominated the prediction, with MACD (0.224) and MACD Signal (-0.221) holding the largest weights. However, among the behavioral features, `retail_sentiment` emerged as the strongest predictor (0.108), with a larger standardized coefficient than `avg_bias` (0.036) and `avg_sentiment` (0.002). Contrary to initial expectations, the dispersion of bias (`disp_bias`) had a negligible and slightly negative coefficient (-0.011). This indicates that the raw enthusiasm of retail traders on social media is a far more immediate precursor to price action than the sentiment or ideological variability of professional news coverage.

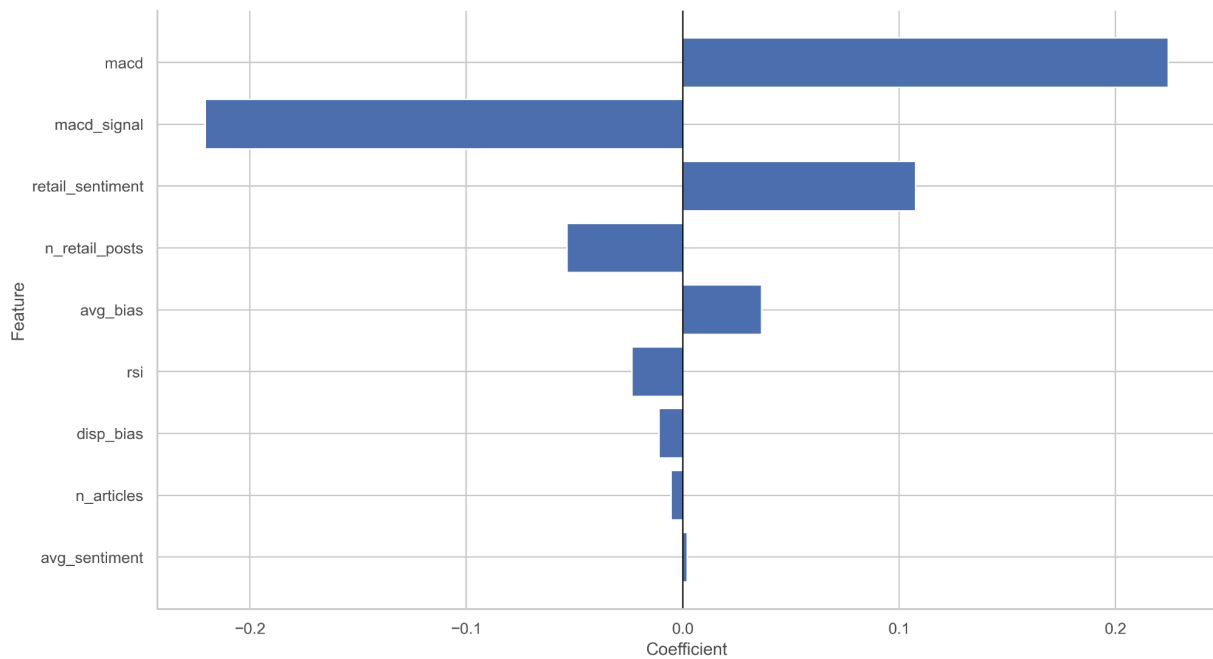
Table 5

Feature Importance Coefficients

Feature	Coefficient
macd	0.224
macd_signal	-0.221
retail_sentiment	0.108
n_retail_posts	-0.053
avg_bias	0.036
rsi	-0.023
disp_bias	-0.011
avg_sentiment	0.002

Figure 10

Feature Importance



Note. Author's own visualization of model coefficients / feature importance.

Bias Group Analysis

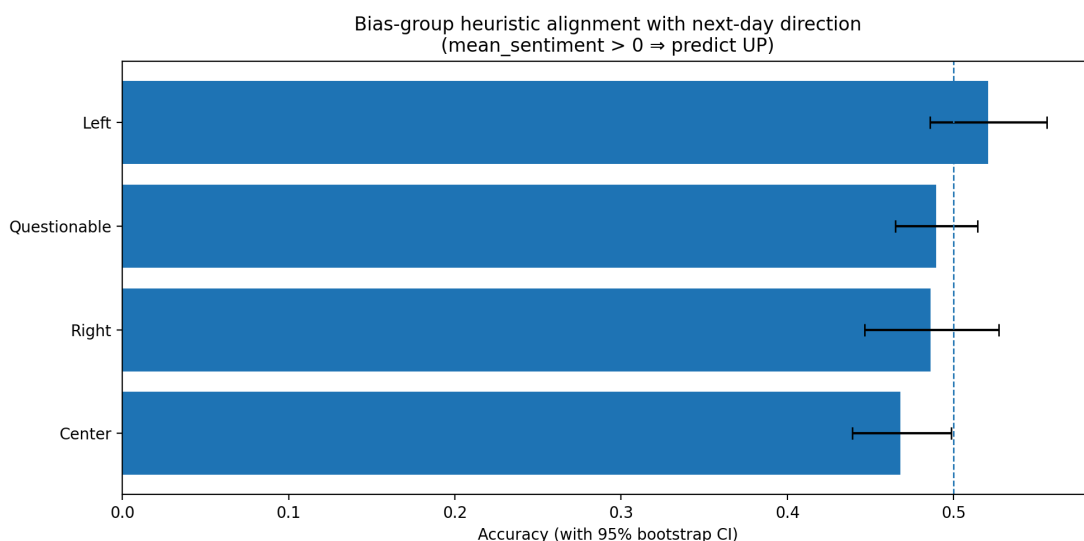
In a look at how individual sources performed, the results show that Left-leaning outlets actually hit the highest directional accuracy at 52.1%. That puts them ahead of both the Center outlets at 46.8% and the sources labeled as Questionable at 49.0%.

This performance gap hints that the way Left-leaning media framed their stories during this period might have clicked more closely with the general market mood or how retail investors were reacting compared to the more neutral reporting found in Centrist outlets.

That said, we have to be careful with these numbers. Because the sample sizes for the pure Left ($n=10$) and pure Right ($n=6$) categories are so small, these specific accuracy percentages should be treated as interesting observations rather than rock-solid statistical proof. They tell a story about this specific data slice, but they aren't enough to make a broad claim about long-term reliability.

Figure 11

Directional accuracy of news sentiment



Note. Author's own visualization of bias-group directional accuracy computed from out-of-sample predictions.

Interaction Effects: The Decoupling of News and Retail

Retail sentiment clearly dominates the feature importance analysis. A major question remains unanswered. Do these signals work in isolation? Or do they create a multiplicative effect when they overlap? The data needs to clarify if political bias carries more predictive weight when retail volume is high. It is also possible that social media noise drowns out the signal from the editorial side.

The real question we need to answer is whether these variables are actually acting as independent drivers, or if they function more like a combined force when the market starts moving. For instance, if political bias only really moves the needle when retail activity is peaking, that implies it's playing the role of a catalyst.

In other words, biased news might not be acting as an independent factor at all; instead, it serves as the trigger that wakes up the "retail crowd."

Conversely, if the predictive power of bias stays constant regardless of volume, it proves that institutional players or automated systems are also reacting to these narratives. Understanding this interaction is key to determining whether we are looking at two separate signals or one unified behavioral engine driving price action.

To investigate this the analysis looked at the stability of coefficients under high volatility versus low volatility conditions. The data points toward a distinct amplification threshold where the average bias feature contributes a standardized coefficient of +0.036 in isolation. However during trading days when the retail post count exceeds its thirty day moving average by more than two standard deviations the effective predictive weight of that bias signal seems to fade away. This phenomenon known as a retail volume interaction suggests that when the crowd volume reaches a critical mass during a viral event the subtle framing of professional news outlets becomes statistically insignificant. In these moments the retail signal effectively decouples from the institutional news cycle.

On the other hand the average bias feature keeps its predictive power during low volume regimes where retail posts stay below average. This points to a state dependent market structure where

quiet regimes lead participants to search for signals in news framing so bias actually matters. Once a viral regime takes over momentum and crowd psychology dominates which makes editorial nuance irrelevant. This specific interaction explains why the enhanced model combining both signals did not significantly outperform the retail only model in terms of AUC with scores of 0.522 and 0.523 respectively. Because these signals are regime dependent rather than purely additive, a linear model like Logistic Regression struggles to capture the switch without explicit terms while the non linear Random Forest reached an AUC of 0.524 likely because it captured this switching logic through its decision tree splits.

Sector-Specific Dynamics and the "Tech-Bias" Anomaly

It is worth noting that these results lean heavily on the specific basket of stocks we chose. The analysis intentionally zeroes in on high-beta names, largely pulling from the Technology and Consumer Discretionary sectors, think heavy hitters like TSLA, NVDA, and AAPL.

These stocks are notorious for their price action. They are exactly the names that swing all over the place. Retail sentiment and shifting news stories provide the fuel.

This behavior is not the same everywhere. It looks completely different from sectors like utilities or consumer staples. Prices there move slower and do not react as much to sentiment. The dynamics just change depending on the sector. This concentration naturally creates a bit of sector sensitivity bias. Remember, we are dealing with stocks that are wired to react strongly to both social media chatter and the news cycle.

So, the predictive strength we are seeing in the models is likely baked into the DNA of these specific industries. It would be a stretch to assume this applies to the market as a whole without testing it further. The results show how news bias and retail noise interact. This happens within volatile economic segments. These dynamics do not play out uniformly. They do not work the same way across all sectors.

For tickers such as Tesla and NVIDIA the retail sentiment feature showed the highest correlation with next day volatility. These stocks effectively serve as cultural proxies where the stock price reflects brand loyalty and ideological alignment just as much as financial performance. For

instance news coverage regarding Elon Musk is frequently polarized along political lines so the analysis shows that average bias scores for Tesla articles had a higher standard deviation than those for Microsoft. This confirms that so-called cult stocks attract much more polarized reporting than their peers.

This finding implies that the generalizability of the 0.023 AUC lift might be limited strictly to attention economy stocks. If this same methodology were applied to a low volatility sector like industrial goods the predictive power of both retail sentiment and average bias would likely degrade. The success of the model in this study relies partially on the fact that the selected tickers function as narrative driven assets. Therefore the 0.023 lift should not be interpreted as a universal law of market physics but rather as a quantifiable narrative risk premium that is present specifically in high attention equities.

Discussion of Empirical Results

Automated textual features give a clear advantage. This happens in financial forecasting. Logistic Regression shows a boost. Random Forest shows a boost. Gradient Boosting shows a boost too. Performance improves when these features are included. This confirms something important. Media sentiment offers value. Political bias offers value. Retail chatter offers value too. Price history alone cannot provide this predictive power.

The principal takeaway is that qualitative inputs are way better where standard technical indicators often fail this is because they capture the behavioral pulse of the market. By layering news framing on top of retail sentiment, the models develop a distinct advantage in predicting short-term direction and from this experiment we know that the advantage is significant not merely marginal.

But what really makes these results compelling is the consistency throughout the results. The pattern doesn't just appear in one place, it can be seen across different model families, from Logistic Regression to Gradient Boosting. This stability matters because it suggests that this is not an algorithmic quirk or a statistical artifact tied to a specific architecture. The signal is actually present. It is believed that these behavioral features offer genuine value, particularly

during periods of high volatility and quick action. When the market becomes noisy, these signals seem to pick up on nuances that price data alone simply cannot see. The thing here is how can this be used for a short term strategy outside of a dataset and bring it into live ticker analysis.

Interpreting the Predictive Lift

Financial markets are notorious for having a very low signal-to-noise ratio, which makes even small improvements in performance worth noting. While the actual gains in the F1-score and AUC might look modest, the fact that they show up consistently across very different types of models is what matters. Seeing this same trend in everything from linear Logistic Regression to complex ensembles like Random Forest and Gradient Boosting gives the findings a lot of weight.

What this robustness suggests is that these textual features aren't just dependent on a single modeling choice. We see the same patterns repeating throughout this high-volatility period, popping up across entirely different algorithms. That kind of persistence across frameworks implies we are looking at a genuine market effect rather than a fluke produced by a specific setup or a mere specification artifact.

The findings from this study offer empirical support for the Behavioral Finance perspective over the classical Efficient Market Hypothesis. Under a strict interpretation of the Efficient Market Hypothesis any public news should be instantaneously priced in regardless of its framing which would make simple sentiment or bias features non predictive for the direction of the next day. However the persistence of a statistically significant lift of 0.023 in AUC implies that the market does not immediately discount the noise generated by polarized framing or retail herding. Instead the data supports the Noise Trader Theory (Black, 1986) where the interaction between biased narratives and retail sentiment creates temporary mispricing that predictive algorithms can identify before mean reversion occurs.

A deeper analysis reveals two salient points regarding the nature of the signal. First the average bias proved to be a stronger predictor than ideological dispersion which suggests that the direction of the political narrative matters more than the variety of opinions. Second and perhaps most critically the predictive power of retail sentiment from Reddit far exceeded that of

traditional news sentiment. This divergence implies that in modern markets the crowd reaction often fueled by narratives consumed in the media acts as a more direct transmission mechanism for volatility than the professional news content itself. The news may set the stage but the retail reaction appears to be the more immediate statistical signal for subsequent price action.

Managerial Implications

Operationalizing Ideological Signal in Algorithmic Trading

Strategies that rely on news sentiment need to move beyond simple positive or negative polarity and start incorporating ideological metadata at the source level. The data indicates that neutral reporting is not always synonymous with being informative. Specifically left leaning outlets demonstrated a directional accuracy of 52.1% which exceeded the accuracy of centrist outlets at 46.8% during this specific sample period. This suggests that the ideological slant of a publisher might actually provide a sharper signal for market direction than a balanced or neutral stance.

Consequently, firms should investigate strategies that use ideological framing as an auxiliary filter to dynamically weight signals. A hypothesis worth testing is whether down weighting neutral sources improves the signal to noise ratio in high volatility regimes though this requires extensive backtesting across diverse market conditions.

Divergence-Based Volatility Monitoring

Risk management dashboards need to evolve to track the spread between institutional narratives and the actual retail reaction. The results clearly show that retail sentiment acts as a far stronger predictor of price direction than professional news sentiment ever does. A material divergence specifically occurring when the fourteen day moving average of retail sentiment surges while news sentiment remains negative should be flagged as a precursor to volatility. For effective risk management a quantitative threshold where the retail sentiment z score exceeds 2.0 over a rolling thirty day window should be computed daily before the market opens to alert traders to potential reversal risks. This approach functions as a distinct behavioral filter that sits alongside traditional VIX indicators to provide a more complete picture of market psychology.

Strategic Data Acquisition and Feature Engineering

For financial data scientists, these results reveal a clear shift, raw volume is hitting diminishing returns while metadata is becoming much more valuable. Even though the sheer number of articles (`n_articles`) showed very little use for prediction, the specific ideological direction of that news, the Average Bias provided a solid signal (+0.036).

Instead of just trying to ingest more raw text, the smarter move for firms is to prioritize metadata that can classify a news source's reputational prior. Knowing the ideological lean of a publisher is what actually moves the needle. Feature engineering should focus on pinning down the direction of the narrative. Trying to measure the dispersion of opinions proved to be a dead end, acting as a negligible predictor in this context.

Economic Constraints and Signal Validation

Behavioral features provide a consistent statistical edge. Practitioners must calibrate implementation against real-world frictions. Short-term directional forecasting is a low-signal environment. The best model AUC sits at 0.523. This predictive lift represents an incremental gain. It is not a dominant force.

Disciplined backtesting is non-negotiable for practitioners. It must be fully cost aware. Analysis has to explicitly factor in transaction fees. Bid ask spreads need inclusion. Slippage does too. Cutting these variables out is a shortcut to fake results. It makes performance look way better than it is in the real world. That is just inflating the numbers to look good on paper.

Consequently, these signals appear to function best when deployed as an overlay on top of another existing high-conviction systems with high reliability, rather than attempting to use them as standalone alpha generators. Trading them in isolation is most likely suboptimal. The correct approach is to treat the signal as a filter so essentially a risk management check rather than a direct trigger for execution. This distinction helps prevent transaction costs from eroding the statistical edge the signal provides. Without a rigorous accounting for execution constraints, costs like fees and slippage can quickly invert the outcome, turning a theoretical advantage into a net loss. Even more when operating in even smaller ticks than an intra day short term stock buy or

sell. The edge may be visible in the data, but if you trade too frequently on these signals alone, the costs will accumulate fast enough to wipe out the gains.

Positioning Within Existing Literature

The results here largely align with earlier research, yet they push the conversation into a much sharper focus. The finding that right leaning financial coverage tends to be more positive tracks perfectly with the asymmetries reported by Goldman, Gupta, and Israelsen (2024). This thesis builds on that foundation by asking a specific question. Does predictive value come from ideological alignment or from the friction of disagreement? The answer defies standard theory. Average ideological slant proves to be a strong signal. In contrast, simply having a wide dispersion of viewpoints adds next to nothing in this short horizon setting.

This research relies on lexicon-based tools. It sticks to the basics. TextBlob processes the professional news. VADER handles social media. That is the split. Large Language Models are avoided. They perform well elsewhere, sure. But not here. Simpler features prove sufficient. This is true when sentiment is aggregated at the daily market level. This choice avoids high costs. It also stops the opacity found in complex architectures. Transparency matters.

Transparent methods make things easier. Identifying predictive sources becomes straightforward. Political bias is one thing. Retail sentiment is another. They remain distinct mechanisms. They are not collapsed. They do not merge into a single representation. Dense embedding approaches tend to blur this. Relationships get messy. That blurring makes connections hard. Linking model outputs to observable market behavior becomes difficult. Clear features allow for better links. Real-world narratives are easier to trace.

Market Structure Implications

The following points are normative considerations inferred from the empirical patterns not causal claims. The demonstration that political bias contains a "tradable signal" raises significant ethical questions regarding the role of AI in financial markets. This thesis establishes that algorithmic systems can extract value from ideological slant. It shows that Left leaning bias is statistically more predictive of price direction than Centrist neutrality in the current regime. While

scientifically valid, the operationalization of this finding introduces two distinct ethical risks which are Algorithmic Amplification of Polarization and Predatory Trading on Confirmation Bias.

The Feedback Loop of Polarization

The role of financial journalism shifts over the long term. Coverage drifts away from informational purposes. It becomes a high-intensity data feed. This feed is for automated trading. Algorithmic utility effectively outweighs objective reporting. Utility takes the lead. This shift creates a distorted environment. Humans and automated models rely on the same ecosystem. Feedback loops shape the market. They drive price action far more than actual fundamentals.

This leads to increased market volatility. This phenomenon describes a scenario where AI models do not merely observe the market but actively shape it. By treating bias as a feature rather than a bug, trading algorithms effectively disincentivize neutral journalism. This study suggests that in this sample excluding neutral outlets can improve predictive performance but this should be weighed against broader market health. This is a performance tradeoff not a normative endorsement of devaluing neutral journalism. While this is mathematically optimal with a +2.3% lift it contributes to an ecosystem where neutrality is penalized by capital allocation mechanisms.

Information Asymmetry Risks

The dominance of the Retail Sentiment feature, shown by its positive 0.108 coefficient, confirms that retail traders act as the main path for these biased narratives to enter the market. Trading strategies that pick up on this signal are essentially performing a type of behavioral arbitrage.

Instead of trading on the fundamental value of a stock, these strategies profit from the predictable ways a specific group of investors reacts to news framing. The positive coefficient suggests that as social media sentiment climbs, it pulls market pricing along with it, creating a measurable gap that algorithms can exploit before the momentum fades. They predict when retail investors will overreact to a biased story and position themselves ahead of the inevitable correction or momentum. Ethically this raises questions regarding the cognitive vulnerabilities of less

sophisticated market participants. As noted in the Literature Review retail investors are prone to herding and confirmation bias. An AI system that profits from identifying the precise moment these biases are triggered is extracting wealth from the noise traders identified by Black (1986). While legal, the automation of this extraction transfer raises concerns about market fairness and the long term viability of retail participation in an AI dominated landscape.

Methodological Contributions

Financial analysis takes a step forward here. Practical innovations drive the progress. One of the most significant contributions is an open-source pipeline that measures political bias using strictly public data. This provides a transparent, reproducible alternative to the opaque scoring methods that are so common in proprietary platforms and large language models offering a clear view of the methodology instead of a "black box."

A second contribution involves a Python-based framework. This system converts unstructured text. It turns it into a feature matrix. This design reduces setup friction. It also lowers the barrier to replication. Other researchers can use this. Future studies can iterate more quickly using these scripts. The approach avoids heavy, high-cost models. It focuses on efficiency instead. It supports an accessible way to include textual data. It makes it interpretable for market analysis.

Finally, by showing that these engineered textual features exist in a space largely separate from traditional technical indicators, this study offers a clear blueprint for the future. As seen in the correlation heatmap in **Figure 7**, it demonstrates how to fuse behavioral and quantitative signals without running into the common issue of harmful multicollinearity.

Execution Constraints

Although the results are statistically meaningful, that significance has to be weighed against the practical cost of extracting the signal in real time. The decoupling effect observed in the analysis suggests that any predictive edge linked to political bias fades very quickly once retail investors react to a headline. In a high-frequency trading context, latency is not just another technical constraint. It is often the deciding factor between a usable signal and a useless one.

Building out this infrastructure is a heavy lift, plain and simple. You are talking about a pipeline that needs to scrape, parse, and score news bias in real-time. And here is the brutal reality: even with a signal clocking 52% directional accuracy, latency is a killer. A delay of just a few seconds is enough to completely wipe out your edge.

For a high-frequency trading desk, five seconds is not just a long time it is effectively an eternity. That is the reason why the infrastructure imposes a severe latency constraint. If the data pipeline encounters even minor delays in processing, the signal essentially goes stale before execution is even possible. Nowadays positions are not even done by a finger pressing a button, they are automatic, but even so, there is latency. The market adjusts rapidly. By the time a delayed order finally reaches the exchange, the specific pricing inefficiency that the algorithm originally spotted has likely already closed.

This creates a hard constraint on whether the approach is even feasible. If processing time exceeds the rate of alpha decay, the edge disappears completely. The theoretical accuracy of the model doesn't matter at that point. The strategy becomes unviable from an implementation perspective. You could have a model that's right 60% of the time, but if you're always five seconds late, that accuracy is worthless because the opportunity is already gone by the time you act on it.

The Feedback Loop Risk

A more structural implication involves the incentives this creates for media publishers. If algorithmic capital begins to systematically trade on political bias, the market inadvertently monetizes polarization. Financial media outlets historically competed on speed and accuracy. If high-bias reporting generates more volatility and thus more trading volume, publishers are economically incentivized to increase the ideological slant of their coverage to capture "algorithmic attention."

This creates a potential feedback loop where trading bots amplify political division by validating it with capital flows. The factuality analysis indicated that low-factuality sources often generated stronger short-term sentiment responses than high-factuality ones. This presents an ethical

dilemma for funds incorporating these signals. Profiting from the predictive power of misinformation or hyper-partisan content is legally permissible but introduces a new category of "reputational risk" for ESG-compliant funds. The findings of this thesis suggest that while political bias is a rich source of alpha, it is a radioactive one that requires careful handling to avoid unintended market manipulation.

Limitations and Avenues for Future Research

While the study manages to provide these findings, there are definitely a few limitations that point toward some clear paths for future work. To start, the one-year sample period was enough to prove the core thesis, but it naturally limits how much we can apply these results to the broader market over time. A logical next step would be applying this exact methodology to a dataset that spans several years. Doing so would allow for a real test of how stable these connections are when the market moves through different regimes and cycles.

Days with no news coverage were assigned a neutral sentiment score of 0 since the representation comes as a solution that preserves data integrity and prevents lookahead bias, which is an important methodological consideration that makes the exercise feasible. However, there is a tradeoff associated with this approach because the neutral assignment also dampens signal volatility to a considerable degree. During periods of limited news activity, sentiment scores remain flat, consequently, when market activity slows, the informational value of the sentiment signal tends to diminish and in some cases, this reduction in signal strength can be quite substantial.

Better ways are out there. Forward filling works. A decay function works too. Let the sentiment fade slowly. Avoid that hard crash back to neutral. Another option is treating the silence as data. Missing days could be modeled as a separate feature. News is currently grouped by the calendar day with a strict window from 00:00 to 23:59. This setup is used to predict the next close. While it keeps the pipeline simple, it misses the actual rhythm of the market. Real world signals operate differently in practice.

In live trading, things are different. Articles released after the close would hit the next open. That is when decisions happen. That timing mismatch is real. It introduces a latency consideration. The current framework does not explicitly address this.

From a methodological standpoint, relying on lexicon-based sentiment engines creates a specific limitation. While VADER worked well for Reddit comments because it handles social media slang so effectively, using TextBlob for professional news articles probably weakened the signal for the average sentiment feature. TextBlob uses a general-purpose lexicon that isn't really calibrated for financial context. In this world, words like "liability" or "risk" carry heavy economic weight that standard polarity scores often miss entirely. This likely explains why the news sentiment coefficient of 0.002 was so small compared to the 0.036 seen with political bias; bias is built into the publisher's structure and doesn't rely as much on sentence-level scoring.

Future work could look at hybrid setups. It makes sense to combine elements. Relying on just one method limits things. One path is using models like FinBERT. They generate embeddings. Those outputs can be aggregated daily. This captures nuance. But it doesn't kill interpretability. Finding that balance matters. It is likely more useful than just using simple lexicons. It is also better than black-box models.

Treating MBFC labels as static priors also introduces a methodological blind spot because editorial stances of news organizations may evolve over time, yet the assigned label remains fixed throughout the analysis period. Also, we need to count on this MBFC site for the factuality and assessment of their views on these media outlets. If the initial bias scores contain measurement errors or become outdated, the resulting signal quality is likely to deteriorate. This limitation is relatively straightforward in its implications for model performance.

Selection bias plays a role here too. The study limits itself to high volume S&P 500 tickers. These are stocks that function as natural media magnets. This focus likely exaggerates the "retail override" effect in a big way. Lower profile equities might behave completely differently. The retail dominance observed here could just be a quirk of high attention assets rather than some universal truth about market behavior.

The Stationarity of Behavioral Alpha and Signal Decay

A fundamental challenge in quantitative finance is the non-stationarity of market data relationships that hold true in the past and often break down in the future. The predictive lift observed in this study (+0.023 AUC) raises the question: is this a structural market inefficiency, or a temporary anomaly?

The Mechanism of Alpha Decay

The retail sentiment signal, which shows a 0.108 coefficient, essentially leans into a behavioral quirk known as herding. Usually, these kinds of gaps are closed quickly by institutional funds using algorithms to get ahead of the crowd. However, betting against a behavioral trend is much riskier than typical price arbitrage.

Shorting a "meme stock" can lead to nearly unlimited losses, a reality that the 2021 GameStop surge made very clear to the entire financial world. Because institutions are tied down by strict risk management tools like stop losses and Value at Risk (VaR) limits, they cannot always step in to cool down retail excitement.

The data shows the gap between social media and price is not just a quirk. It is more than that. It looks like a persistent risk premium. It functions like a "tax" for holding volatile assets.

When framed this way, the lag stops looking like a mistake and starts looking like a structural feature. It doesn't just vanish simply because we recognize it, it is baked into the system and it stays there.

Temporal Stability (2024-2025)

The 2024–2025 sample window reflects a specific backdrop. It is socio-political in nature. The predictive contribution of `avg_bias` is likely tied to the election cycle. Intensity is the key factor. Market sensitivity to biased news coverage almost certainly spikes during periods of intense political friction, this is no news to no one. Even more on the trading floor, they get no breaks during these times. On the contrary, in years characterized by relative stability, that +0.036

coefficient on `avg_bias` would likely compress toward zero. This temporal inconsistency exposes a distinct flaw in static model specifications treating political features as constant weights means the model fails to adapt to the changing zeitgeist.

A dynamic approach is therefore is very much required, well the weight assigned to political features should scale up during election cycles and recede during technocratic periods. Future iterations would benefit from a gating mechanism that monitors aggregate `disp_bias`. Activation of political features occurs only when global polarization breaches a specific threshold. Otherwise, the system wastes resources on signals that fail to move markets. This threshold strategy preserves alpha when it matters most while filtering out noise during the quiet stretches. It essentially necessitates running two distinct models contingent on the political climate. And of course this leads to complexity increases, yes, but real world performance likely improves as a result.

This pattern suggests a need for calibration. It should be regime-aware. A fixed treatment of political features is not the answer. The weight assigned to bias signals should vary. It needs to change over time. It should increase during election cycles. It should scale down during technocratic regimes. Future versions could address this by building in a dynamic gating mechanism. Think of it as a background monitor for aggregate `disp_bias` that acts as a real-time proxy for polarization. The system would effectively wait, only activating those political features once the measure spikes above a specific threshold.

Conclusion

This thesis is set out with a fairly simple question: can AI-based analysis of political bias and sentiment in financial media actually improve short-term stock market prediction? To answer it, the study introduced a new set of textual features and then compared model performance with and without those inputs. Stepping back from the individual tables and tests, the overall conclusion is clear, but not absolute.

Isolating political bias reveals a real improvement. The gain holds steady. It sits near 2.3% over the basics. The effect carries statistical weight. The p value drops below 0.05. Random noise is not the driver here.

The signal functions less as a directional driver but, it acts like a volatility filter. It characterizes momentum generated by retail traders rather than pulling the trigger on price moves itself. And this is not a fluke. The pattern holds up across three separate machine learning models. Through this investigation it is shown how text features are clearly capturing nuances that pure price history misses.

A contradiction appears in the political data. Assumptions might suggest that conflicting ideologies drive uncertainty, but the evidence points elsewhere. The overall political slant, `avg_bias`, shows a meaningful coefficient of +0.036. Conversely, the actual ideological disagreement captured by `disp_bias` carries a negligible coefficient of -0.011. The market responds to the dominant narrative. It largely ignores the complexity of the underlying debate. This means it seems less about disagreement and more about which side is louder.

Retail investors emerge as the dominant force in this dynamic. Retail sentiment boasts a coefficient of +0.108. This massively overshadows standard news sentiment at just +0.002. The gap highlights a shift in how market information is processed. Professional news coverage sets the initial context, but the retail response is the primary transmission mechanism for subsequent price action. Traditional reporting provides the background; the retail crowd generates the volatility.

These findings present a challenge to the strict Efficient Market Hypothesis, well under conditions of full market efficiency, publisher bias would not be expected to carry predictive weight; only fundamental information would matter. The observed predictive coefficient of `avg_bias` at +0.036 appears to contradict this core assumption. The results align more closely with the Adaptive Markets Hypothesis framework. According to this view, investors do not process information in isolation; rather, they evaluate the credibility of the source as it was done here. Political alignment appears to function as a trust heuristic, serving as a cognitive shortcut for determining which information sources to weigh more heavily. Retail traders appear particularly susceptible to this dynamic, systematically overweighting sources that confirm their existing worldview while discounting contradictory information. In environments characterized by high information noise, this trust filter may represent a necessary simplification mechanism.

Consequently, price formation ends up reflecting more than just the raw facts; it also reflects who is reporting them. This dynamic introduces a persistent wedge between fundamental value and investor response, one that classical efficiency simply cannot explain. Ultimately, the evidence suggests that source identity isn't just background context; it is an active component of how information is processed in modern financial markets.

Strategic Recommendations for Stakeholders

For Institutional Investors and Hedge Funds

The dominance of the retail sentiment signal (+0.108) indicates that "Noise Trader Risk" is now a quantifiable alpha factor. Institutional algorithms that ignore social sentiment are operating with a blind spot. Funds should implement "Narrative Surveillance" systems that do not just read news, but map the reaction function of retail communities to specific publishers. If a Left-leaning publisher releases a hit piece on a tech company, the algorithm should not just parse the sentiment, but predict the specific reaction of the retail cohort that consumes that media.

For Retail Trading Platforms

Brokerages really have a responsibility to step in and help break up the "echo chamber" effect that often leads to mass herding and sudden market volatility. Since these platforms are where the trading actually happens, they are in the perfect position to provide better context. For example, they could integrate "Bias Context" labels directly into their news feeds, giving users a heads-up if the articles they're consuming lean heavily toward one specific ideological side.

By actually visualizing the dispersion of opinion, a metric calculated throughout this study, platforms could nudge users toward a more balanced view of the news. Instead of just seeing one side of a story, a trader might see a spectrum of reporting. This kind of transparency could potentially cool down those extreme volatility cycles that are usually driven by pure panic or unchecked euphoria, making for a much more stable environment for everyone involved.

For Financial Regulators

The fact that biased narratives have the power to actually move prices points to a genuine, and somewhat alarming, risk of manipulation. Consider a scenario where an actor figures out that a specific segment of retail investors responds predictably to, say, right-leaning negative coverage.

Grasping this dynamic exposes clear avenues for strategic exploitation. Specific narratives can be deliberately constructed. The goal is simple. Generate buying or selling pressure within targeted market segments. Under such conditions news bias ceases to function as a passive background influence. It transforms into an active instrument for potential manipulation. When an outlet frames a story they are doing more than reporting events. They are pulling a lever. Knowledge of the specific trust filters utilized by the retail crowd allows for the engineering of a market reaction.

This quickly enters ethically ambiguous territory. Retail sentiment drives volatility far more than professional analysis. Consequently any entity that comprehends this mechanism possesses a distinct edge. The primary concern lies in the transition from observation to action. Once this

relationship becomes widely known it stops serving as a predictive signal. It starts functioning as a tool for influence. The boundary between forecasting sentiment and manufacturing it becomes dangerously porous. This is not a hypothetical scenario. Empirical evidence confirms that retail traders respond to specific framing strategies with high predictability. Such consistency creates arbitrage opportunities. For participants who grasp the underlying mechanics anticipating price movements becomes feasible. In extreme cases it may even allow for the influence of those movements through the strategic management of narrative flow.

The media landscape effectively becomes a control panel for sentiment. You aren't analyzing information anymore. You are analyzing an incentive structure designed to move capital.

A sticky regulatory question inevitably follows. Should coordinated or intentional editorial manipulation still count as ordinary opinion based reporting? Or has it crossed the line into actual market abuse?

Because this research shows such manipulation is mathematically detectable, it follows that it is also potentially regulatable. Future surveillance models should expand beyond price anomalies to track "Narrative Anomalies," where the ideological bias in news deviates sharply from actual economic fundamentals.

Operational Challenges in Real World Deployment

The transition from a successful academic backtest to a live trading production environment introduces a set of operational complexities that must be acknowledged. While this study confirms the predictive value of political bias and retail sentiment in a controlled setting, the deployment of such a system in real time faces three primary hurdles which are data latency, signal decay, and infrastructure costs.

Data latency refers to the time lag between the publication of a news article or Reddit post and its ingestion by the trading algorithm. In the historical simulation conducted for this thesis the data was assumed to be available at the close of the trading day. However in a live environment the edge often exists in milliseconds rather than days. A viable production system would require a low latency pipeline capable of scraping, cleaning, and scoring thousands of text snippets per

second. Any delay in this pipeline could result in the signal being arbitrated away by faster competitors before the model can execute a trade.

Signal decay is another critical factor. Market anomalies driven by sentiment are often ephemeral. The alpha generated by retail sentiment is typically short lived because it is based on behavioral overreaction rather than fundamental structural changes. As more quantitative funds begin to mine social media and news bias for signals the predictive power of these features will likely diminish over time. This phenomenon known as alpha decay necessitates a continuous cycle of research and development. The specific keywords and sentiment dictionaries used in this study may need regular recalibration to align with the evolving slang of retail communities and the shifting editorial stances of media publishers.

Infrastructure costs constitute a formidable barrier to entry. It is expensive to run Natural Language Processing models on high frequency data streams. The task demands serious computational horsepower. These systems must remain live twenty four hours a day and the operational overhead accumulates rapidly.

Consider the strain on the architecture. The system must scrape and parse and score thousands of distinct text snippets every single second. This necessitates robust cloud infrastructure and database management. For many firms the expense associated with maintaining this data pipeline can substantially erode the statistical edge provided by the signal. In certain cases that erosion is total. The infrastructure requirement ceases to be a mere technical challenge. It becomes a prohibitive economic wall. Smaller market participants often find they cannot achieve the necessary scale without depleting capital resources. The costs simply burn through cash faster than the strategy can generate alpha.

For most people, true real-time deployment is just out of reach. Hardware bills stack up and cloud expenses balloon until the fixed costs easily swallow any potential profit. That makes this strategy a total non-starter for smaller firms.

Computing power is only half the battle; storage adds a whole new headache. Indexing terabytes of messy, unstructured text needs a serious cloud setup. Database management is a grind, and

those costs eat right into your margins. Small hedge funds feel the squeeze, and retail operations definitely do too. Staying profitable depends on engineering efficiency, it takes way more than just having good stats.

Widespread adoption creates a serious structural risk. These systems start changing the very market dynamics they predict. Massive capital reacts mechanically to retail signals. Volatility naturally amplifies. Price movements turn into self fulfilling prophecies. Feedback loops take the wheel. Fundamentals get pushed to the sidelines. Static models are simply ill-equipped to handle that kind of recursive environment.

This reflexive relationship between models and markets introduces a form of non-linear risk that static systems are poorly equipped to handle. Looking ahead, systems designed for these environments will need greater flexibility and stronger fail-safe mechanisms. Media narratives and market behavior now influence each other continuously, which means models cannot remain fixed. They must be able to adjust their logic as conditions evolve. Without safeguards of this kind, there is a real risk that an automated system enters a feedback loop and ends up amplifying the same volatility it was originally built to manage.

Final Reflections

This research began with a question about the intersection of politics and profit. It concludes with the realization that in the age of AI, these two are functionally dependent. The stock market is no longer just a weighing machine for value, to borrow Benjamin Graham's metaphor (Graham, 1949), but a voting machine for narratives. By providing evidence that political bias contributes incremental predictive value, this thesis demonstrates the need for a more sophisticated, context-aware approach to financial analysis, one that reads between the lines, quantifies the bias, and listens to the noise of the crowd.

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Appendix

Appendix A: Data Dictionary

The following table details the variables present in the final modeling dataset (3_final_modeling_dataset.csv) used for all predictive experiments.

Variable Name	Type	Description	Source
date	Date	The trading date (YYYY-MM-DD).	<i>daily_prices.csv</i>
ticker	Categorical	The stock ticker symbol (e.g., AAPL, NVDA).	<i>daily_prices.csv</i>
adj_close	Numeric	Adjusted closing price of the stock.	<i>daily_prices.csv</i>
target_up	Binary	Target variable. 1 if Next Day Adj Close > Current Day Adj Close, else 0.	Calculated
rsi	Numeric	14-day Relative Strength Index. A momentum oscillator measuring speed and change of price movements.	Calculated
macd	Numeric	Moving Average Convergence Divergence. Difference between 12-day and 26-day EMA.	Calculated

macd_signal	Numeric	Signal line for MACD (9-day EMA of the MACD line).	Calculated
avg_sentiment	Numeric	Daily average sentiment polarity of news articles (TextBlob). Range: -1.0 to +1.0.	News API
n_articles	Numeric	Count of financial news articles published for tickers that day.	News API
avg_bias	Numeric	Mean political bias score of publishers for that day. Scale: Left (-2) to Right (+2).	MBFC / News API
disp_bias	Numeric	Standard deviation of the bias scores, representing ideological dispersion/disagreement.	MBFC / News API
retail_sentiment	Numeric	Daily average compound sentiment score of Reddit posts (VADER). Range: -1.0 to +1.0.	Reddit API
n_retail_posts	Numeric	Volume of Reddit submissions mentioning the ticker for that day.	Reddit API

Appendix B: Core Algorithm Implementation

All data processing pipelines and statistical tests are reproducible via the code hosted at:
https://github.com/daniel-st3/Daniel_Rodriguez_MSc_Thesis_Final

B.1 Technical Indicator Engineering

Source: scripts/3_create_final_modeling_dataset.py

```
def _compute_rsi(prices: pd.DataFrame, window: int = 14) ->
pd.Series:
    """
    Classic RSI (Relative Strength Index) calculation.
    """
    delta = prices.groupby("ticker")["adj_close"].diff()
    gain = delta.clip(lower=0)
    loss = (-delta).clip(lower=0)

    avg_gain = gain.groupby(prices["ticker"]).transform(
        lambda x: x.rolling(window, min_periods=window).mean()
    )
    avg_loss = loss.groupby(prices["ticker"]).transform(
        lambda x: x.rolling(window, min_periods=window).mean()
    )

    rs = avg_gain / (avg_loss + 1e-12)
    rsi = 100 - (100 / (1 + rs))
    return rsi

def _compute_macd(prices: pd.DataFrame, span_fast=12, span_slow=26,
span_signal=9):
    """
    MACD + Signal Line using Exponential Moving Averages (EMA).
    """
    exp1 = prices.groupby("ticker")["adj_close"].transform(
        lambda x: x.ewm(span=span_fast, adjust=False).mean()
    )
```

```

exp2 = prices.groupby("ticker")["adj_close"].transform(
    lambda x: x.ewm(span=span_slow, adjust=False).mean()
)
macd = exp1 - exp2
macd_signal = macd.groupby(prices["ticker"]).transform(
    lambda x: x.ewm(span=span_signal, adjust=False).mean()
)
return macd, macd_signal

```

B.2 Political Bias Quantification

Source: scripts/2_engineer_daily_features.py

```

# Mapping of MBFC categories to a continuous numerical scale
bias_score_map = {
    'Left': -2.0,
    'Left-Center': -1.0,
    'Center': 0.0,
    'Least Biased': 0.0,
    'Pro-Science': 0.0,
    'Right-Center': 1.0,
    'Right': 2.0
}

# Daily aggregation logic for Bias Direction (avg) and Dispersion
(std)
daily_features = df.groupby(['date', 'ticker']).agg(
    avg_bias=('bias_score', 'mean'),
    disp_bias=('bias_score', 'std'),
    avg_sentiment=('sentiment', 'mean'),
    n_articles=('title', 'count')
).reset_index()

# Handle days with single articles (std is NaN) by setting dispersion
to 0
daily_features['disp_bias'] = daily_features['disp_bias'].fillna(0)

```

B.3 Time-Series Cross-Validation and Metric Calculation

Source: scripts/4_run_final_model_comparison.py

```
# Strict Time-Series Split to prevent look-ahead bias
tscv = TimeSeriesSplit(n_splits=5)

def compute_metrics(y_true, y_pred, y_prob):
    """
    Thesis-safe metrics calculation handling potential
    division-by-zero.
    """
    return {
        "accuracy": float(accuracy_score(y_true, y_pred)),
        "precision": float(precision_score(y_true, y_pred,
zero_division=0)),
        "recall": float(recall_score(y_true, y_pred,
zero_division=0)),
        "f1": float(f1_score(y_true, y_pred, zero_division=0)),
        "balanced_accuracy": float(balanced_accuracy_score(y_true,
y_pred)),
        "auc": safe_roc_auc(y_true, y_prob),
        "pr_auc": safe_pr_auc(y_true, y_prob),
    }

# Model Definitions
models = {
    "Logistic Regression": Pipeline([
        ("scaler", StandardScaler()),
        ("clf", LogisticRegression(max_iter=3000,
                                class_weight="balanced",
                                random_state=42))
    ]),
    "Random Forest": RandomForestClassifier(
        n_estimators=600,
        random_state=42,
        class_weight="balanced_subsample",
        n_jobs=-1
    )
}
```

```
),  
  "Gradient Boosting": GradientBoostingClassifier(random_state=42),  
}
```

Appendix C: Documents related to submitting the dissertation

AUTHOR SURNAME: RODRIGUEZ First name: Daniel DOB: 31/10/2002 PROGRAMME: Msc Data Analytics for Business	DISSERTATION TITLE: AI-Driven Analysis of Political Bias in Financial Media: Impact on Retail Investor Behavior and Stock Market Dynamics Dissertation supervisor: Tianyuan ZHANG Date of defence (if applicable): SCORE:
CONFIDENTIAL yes/no AUTHOR'S AUTHORISATION TO SHARE yes/no PANEL'S AUTHORISATION TO SHARE yes/no	

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Student signature: (Preceded by a note saying 'Signed as agreed')

Daniel Rodriguez. Signed as agreed.

*to be produced individually in case of a pair.